

Lecture 15

Impulse and momentum principle in planar motion

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MC2103
Fall 2026



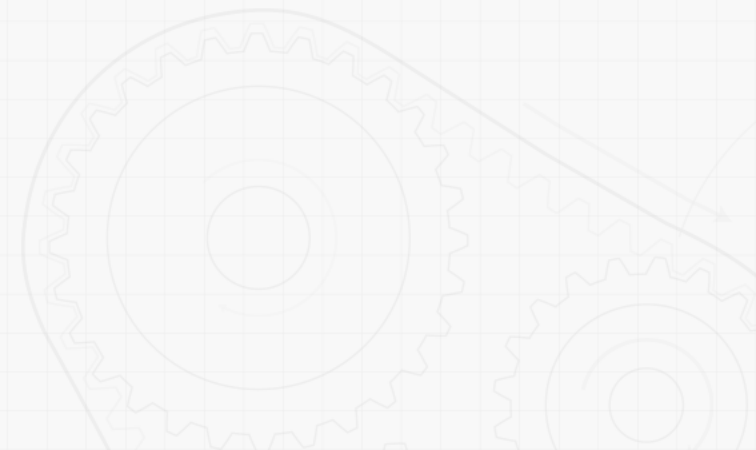
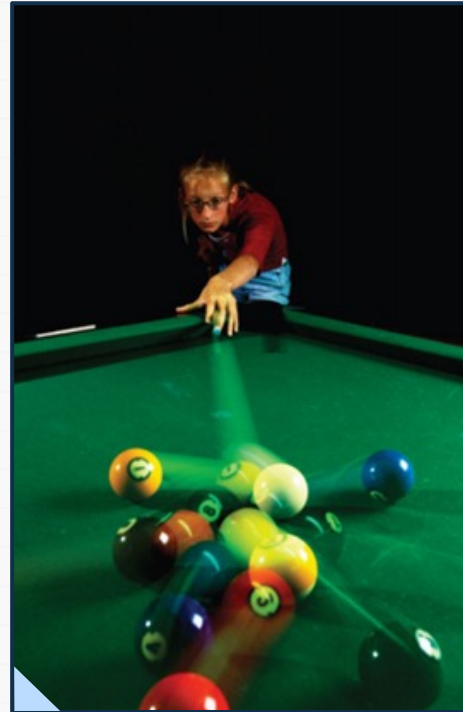
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<https://youtu.be/VO38xJmzxkg>

15 Impulse and momentum principle in planar motion

Angular Impulse Momentum

When two rigid bodies collide, we typically use principles of **angular impulse momentum**. We often also use linear impulse momentum (like we did for particles).



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Introduction

⚙ Approaches to Rigid Body Kinetics Problems

Forces and
Accelerations



Newton's Second
Law(last chapter)

$$\sum \vec{F} = m\vec{a}_G$$
$$\sum \vec{M}_G = \dot{\vec{H}}_G$$

Velocities and
Displacements



Work-Energy

$$T_1 + U_{1 \rightarrow 2} = T_2$$

Velocities and
Time



Impulse-
Momentum

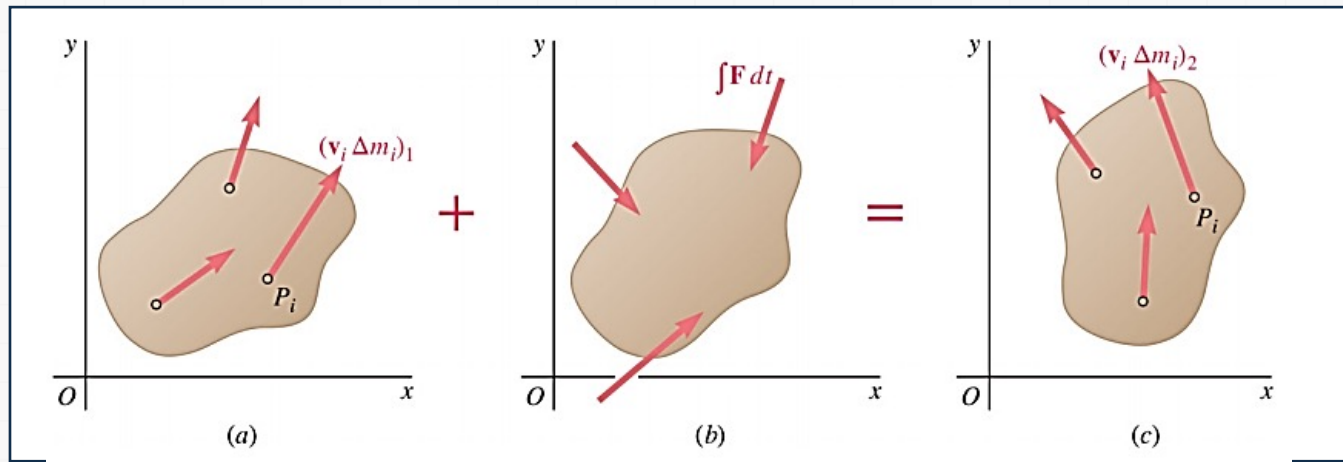
$$m\vec{v}_1 + \int_{t_1}^{t_2} \vec{F} dt = m\vec{v}_2$$
$$I_G \omega_1 + \int_{t_1}^{t_2} M_G dt = I_G \omega_2$$

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Principle of Impulse and Momentum 1

⚙ Method of impulse and momentum

- Well suited to the solution of problems involving time and velocity
- The only practicable method for problems involving impulsive motion and impact.



$$\text{Sys Momenta}_1 + \text{Sys Ext Imp}_{1 \text{ to } 2} = \text{Sys Momenta}_2$$

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Principle of Impulse and Momentum 2

⚙ Method of impulse and momentum

- The momenta of the particles of a system may be reduced to a vector attached to the mass center equal to their sum,

$$\vec{L} = \sum \vec{v}_i \Delta m_i = m \vec{v}$$

and a couple equal to the sum of their moments about the mass center,

$$\vec{H}_G = \sum \vec{r}_i' \times \vec{v}_i \Delta m_i$$

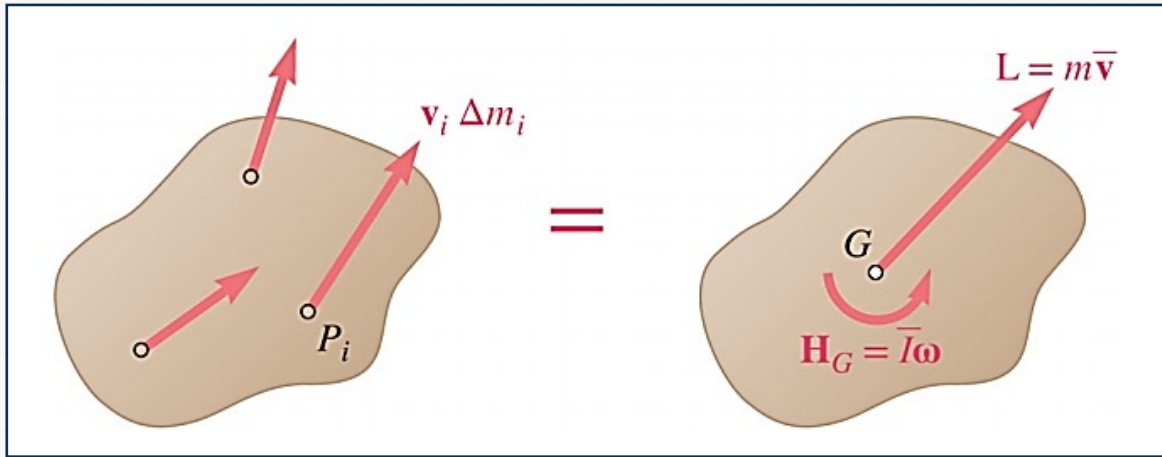
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Principle of Impulse and Momentum 2

⚙ Method of impulse and momentum

- For the plane motion of a rigid slab or of a rigid body symmetrical with respect to the reference plane,

$$\vec{H}_G = \bar{I}\omega$$

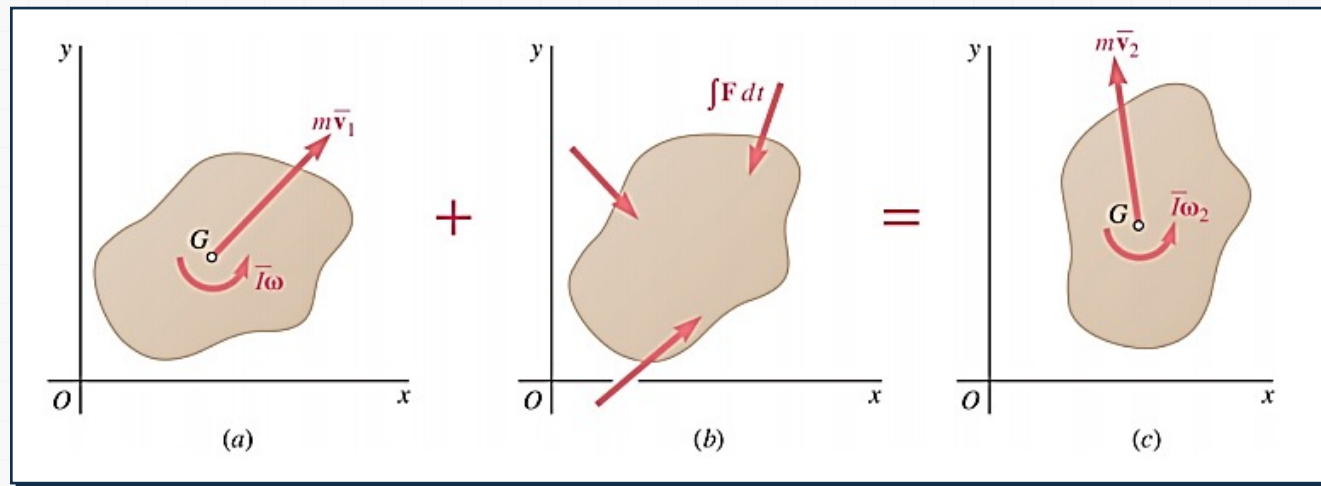


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Principle of Impulse and Momentum 3

⚙ Impulse-momentum diagram

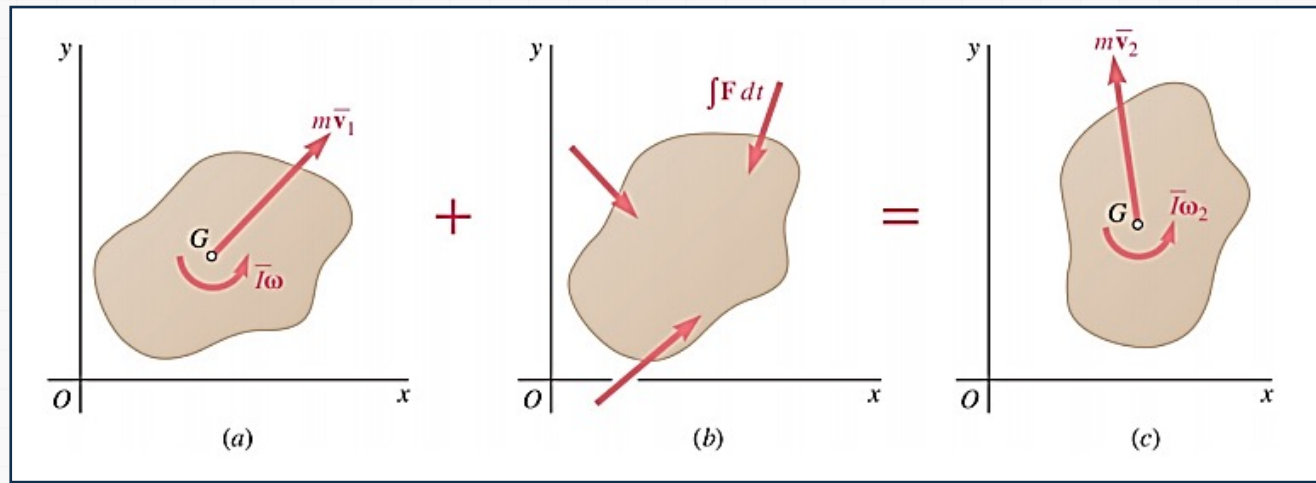
- For plane motion problems, draw out an **impulse-momentum diagram**, (similar to a free-body diagram).



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Principle of Impulse and Momentum 3

⚙ Impulse-momentum diagram



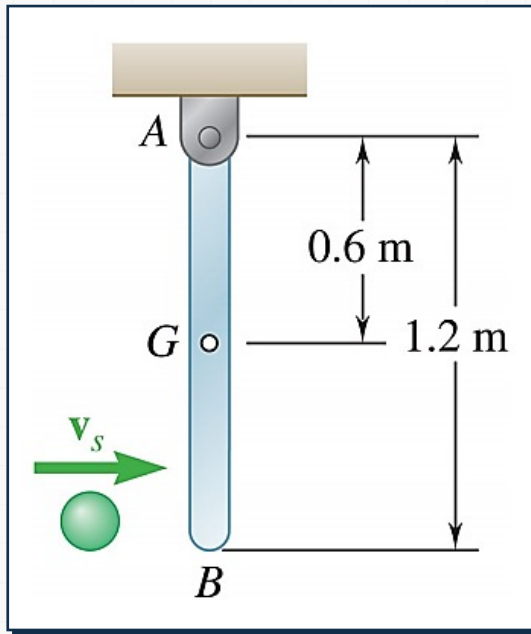
- This leads to three equations of motion.

- Summing and equating momenta and impulses in the x and y directions.
- Summing and equating the moments of the momenta and impulses with respect to any given point (often choose G).

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Impulse Momentum Diagrams 1

⚙ Draw the impulse-momentum diagram



- A sphere S hits a stationary bar AB and sticks to it. Draw the **impulse-momentum diagram** for the ball and bar separately; time 1 is immediately before the impact and time 2 is immediately after the impact.

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Impulse Momentum Diagrams 2

⚙ Draw the impulse-momentum diagram

Momentum of the
ball before impact

Impulse on ball

Momentum of the
ball after impact

$$m_s v_s + F_{imp} \Delta t = m_s v'_s$$

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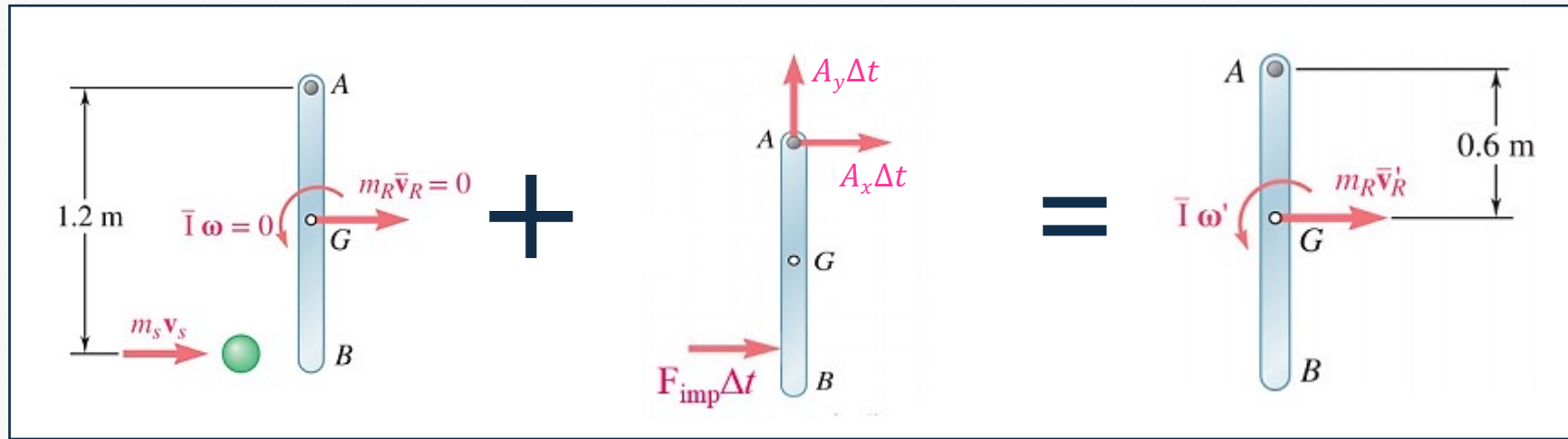
Impulse Momentum Diagrams 2

⚙ Draw the impulse-momentum diagram

Momentum of the bar before impact

Impulse on bar

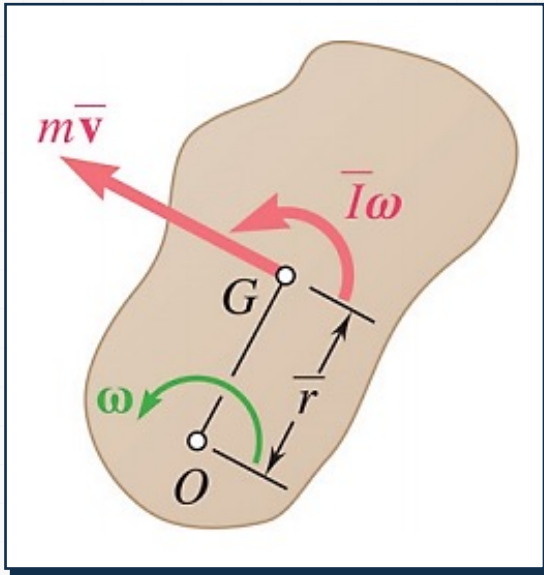
Momentum of the bar after impact



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Principle of Impulse and Momentum

⚙ Fixed axis rotation



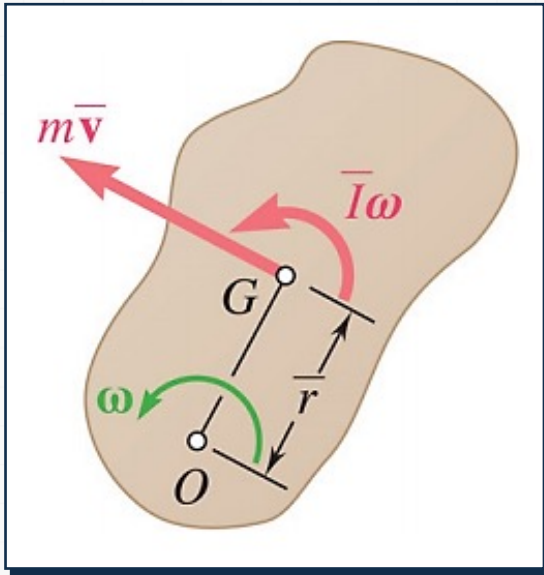
- The angular momentum about O

$$\begin{aligned} I_O \omega &= \bar{I} \omega + (m\bar{v})\bar{r} \\ &= \bar{I} \omega + (m\bar{v} \omega)\bar{r} \\ &= (\bar{I} + m\bar{r}^2) \omega \end{aligned}$$

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Principle of Impulse and Momentum

⚙ Fixed axis rotation



- Equating the moments of the momenta and impulses about O ,

$$I_O\omega_1 + \sum \int_{t_1}^{t_2} M_O dt = I_O\omega_2$$

The pin forces at point O now contribute no moment to the equation.

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Systems of Rigid Bodies

⚙ Rigid Bodies

Motion of several rigid bodies

- Motion of several rigid bodies can be analyzed by applying the principle of impulse and momentum to each body separately.

The principle of impulse and momentum

- For problems involving no more than three unknowns, it may be convenient to apply the principle of impulse and momentum to the system as a whole.

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Systems of Rigid Bodies

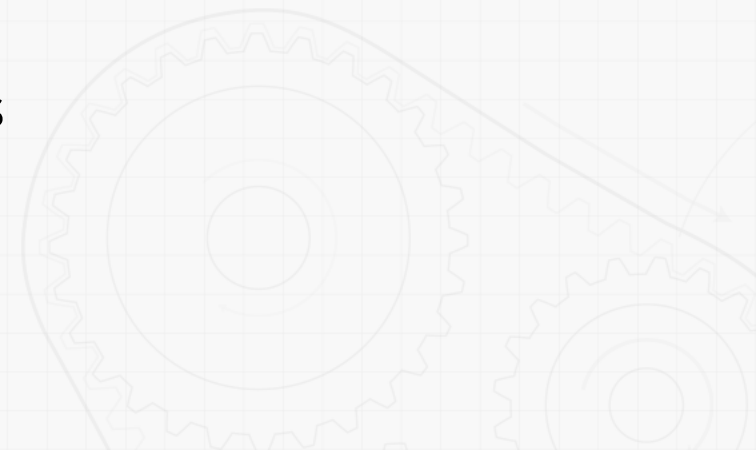
⚙ Rigid Bodies

**Moving
part of
the
system**

- For each moving part of the system, the diagrams of momenta should include a momentum vector and/or a momentum couple.

**Internal
forces**

- Internal forces occur in equal and opposite pairs of vectors and generate impulses that cancel out.



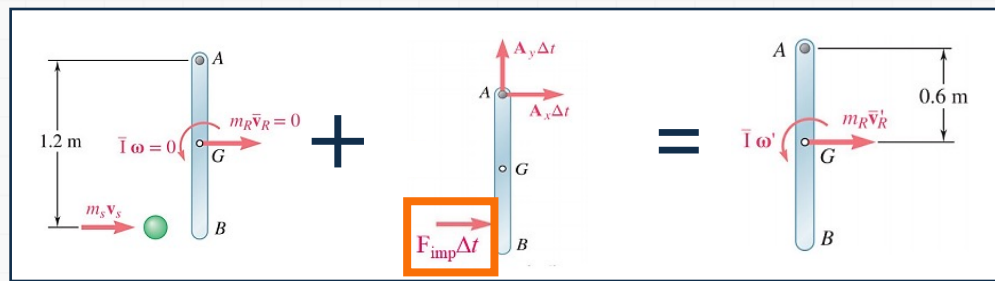
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Practice

⚙ Draw the impulse-momentum diagram

- From the previous problem, notice that the impulse acting on the sphere is equal and opposite to the impulse acting on the bar.

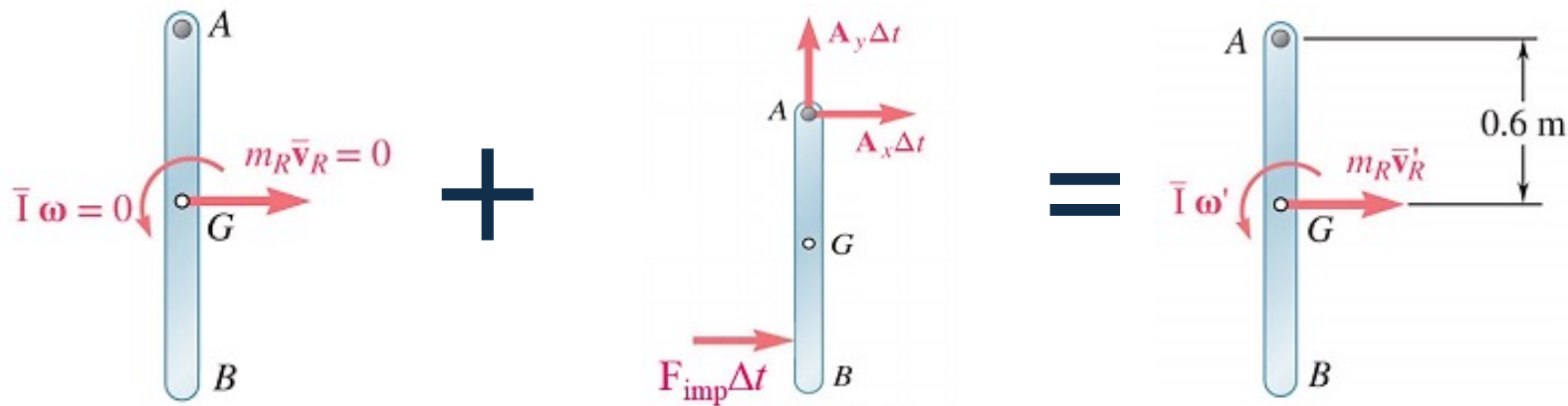
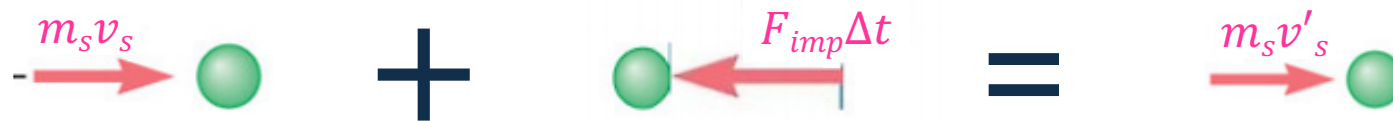
We can take advantage of this by drawing the **impulse-momentum** diagram of the entire system, as shown on the next slide.



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Practice

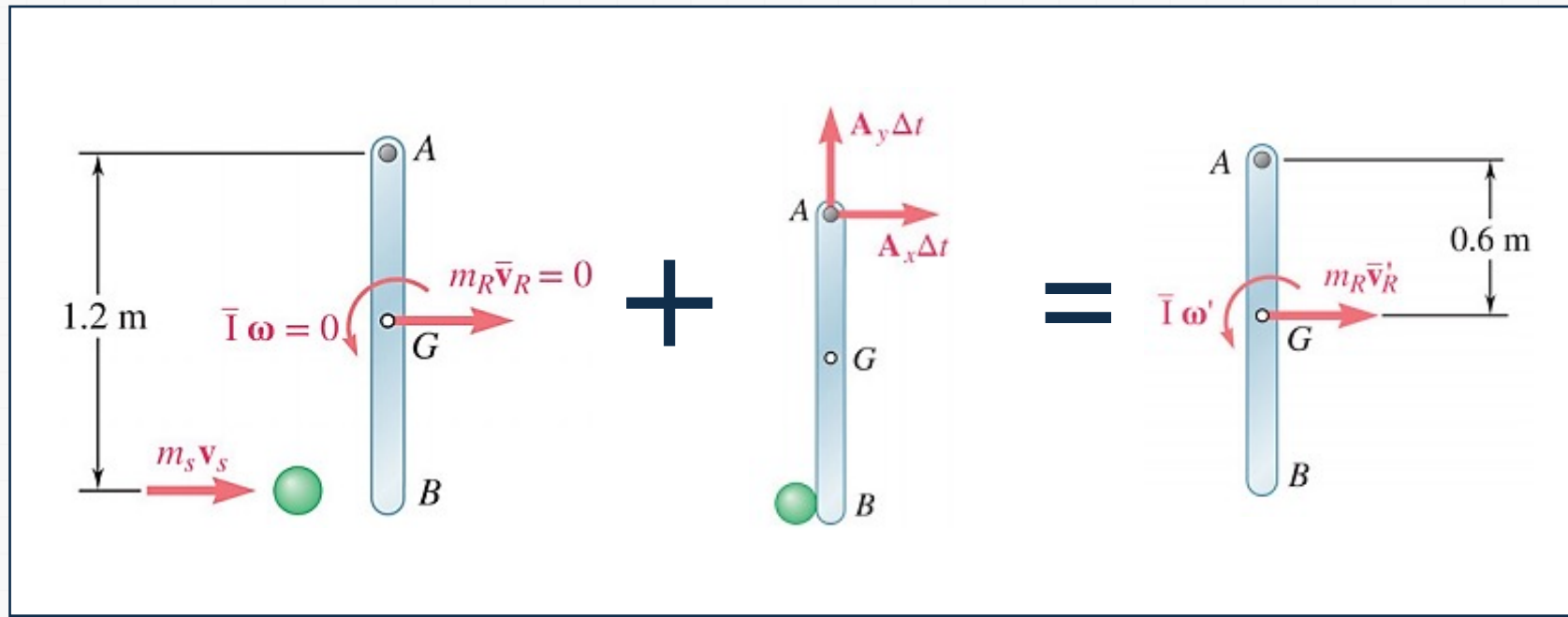
⚙ Diagram for combined system



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Practice

⚙ Diagram for combined system



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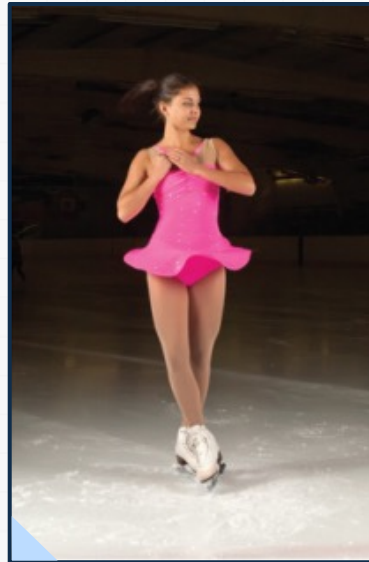
Conservation of Angular Momentum 1

⚙ Diagram for combined system

The moments acting through the skater's center of gravity are negligible, so **her** angular momentum remains constant.
She can adjust **her** spin rate by changing **her** moment of inertia.



$$I_G \omega_1$$

$$=$$


$$I_G \omega_2$$

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Conservation of Angular Momentum 2

⚙ Diagram for combined system

- When no external force acts on a rigid body or a system of rigid bodies, the system of momenta at t_1 is equipollent to the system at t_2 .

The total linear momentum and angular momentum about any point are conserved,

$$\vec{L}_1 = \vec{L}_2 \qquad (H_0)_1 = (H_0)_2$$

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Conservation of Angular Momentum 2

⚙ Diagram for combined system

- When the sum of the angular impulses pass through O , the linear momentum may not be conserved, yet the angular momentum about O is conserved,

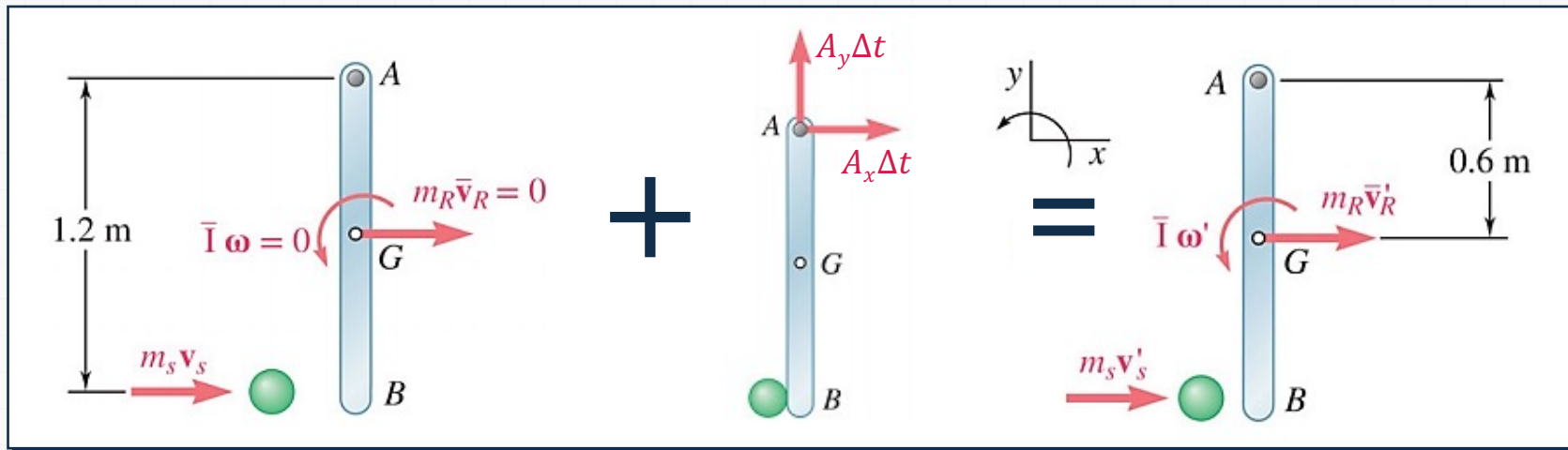
$$(H_0)_1 = (H_0)_2$$

- Two additional equations may be written by summing x and y components of momenta and may be used to determine two unknown linear impulses, such as the impulses of the reaction components at a fixed point.

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Concept Question 1

⚙ Diagram for combined system



For the problem we looked at previously, is the **angular momentum about G** conserved?

Yes

No

For the problem we looked at previously, is the **angular momentum about point A** conserved?

Yes

No

For the problem we looked at previously, is the **linear momentum of the system** conserved?

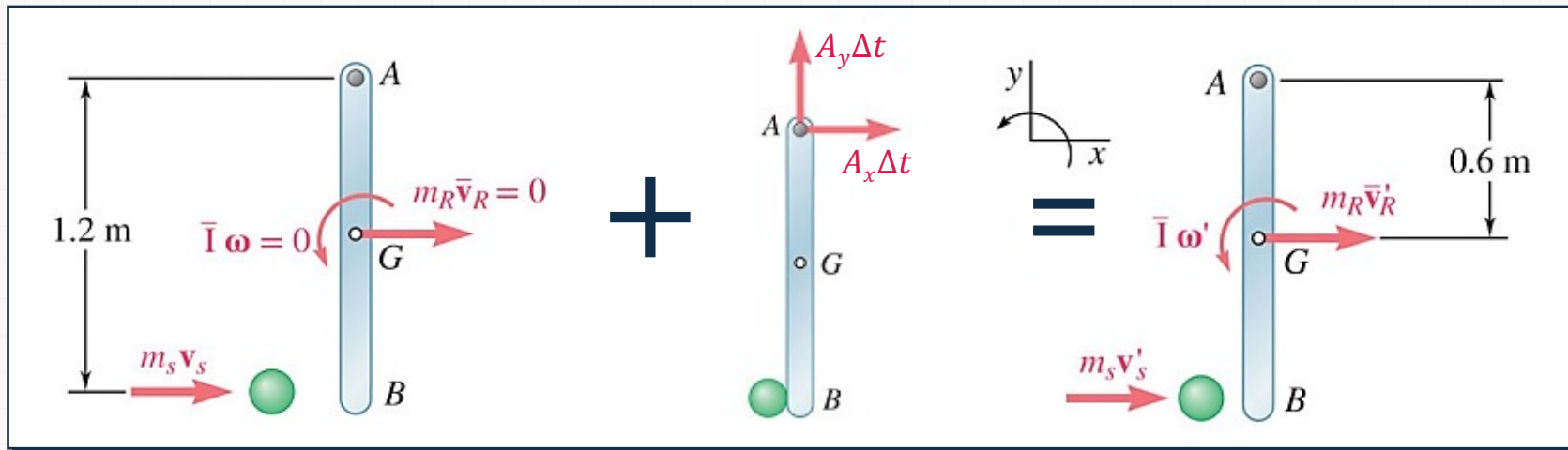
Yes

No

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Concept Question 1

Diagram for combined system



For the problem we looked at previously, is the **angular momentum about G** conserved?

Yes

No

For the problem we looked at previously, is the **angular momentum about point A** conserved?

Yes

No

For the problem we looked at previously, is the **linear momentum of the system** conserved?

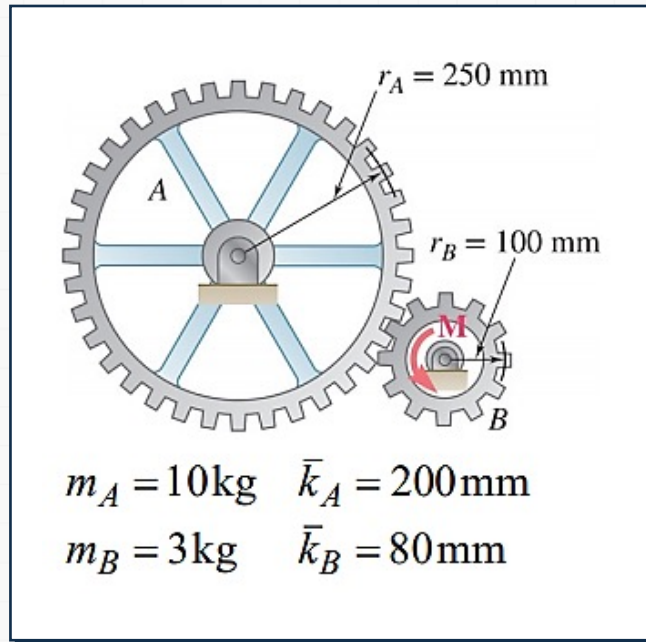
Yes

No

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Sample Problem 1

⚙ Determine the time and the tangential force

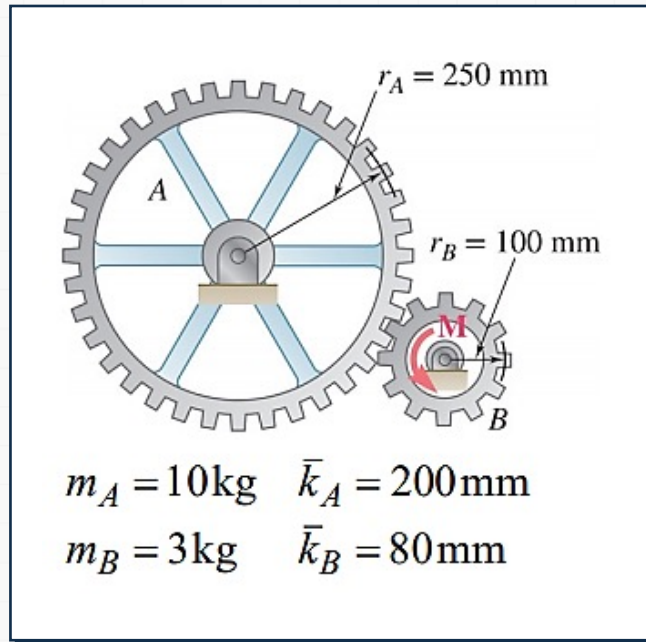


- The system is at rest when a moment of $M = 6 \text{ N} \cdot \text{m}$ is applied to gear B . Neglecting friction,
 - a) determine the time required for gear B to reach an angular velocity of 600 rpm, and
 - b) the tangential force exerted by gear B on gear A .

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Sample Problem 1

⚙️ Determine the time and the tangential force



• Strategy

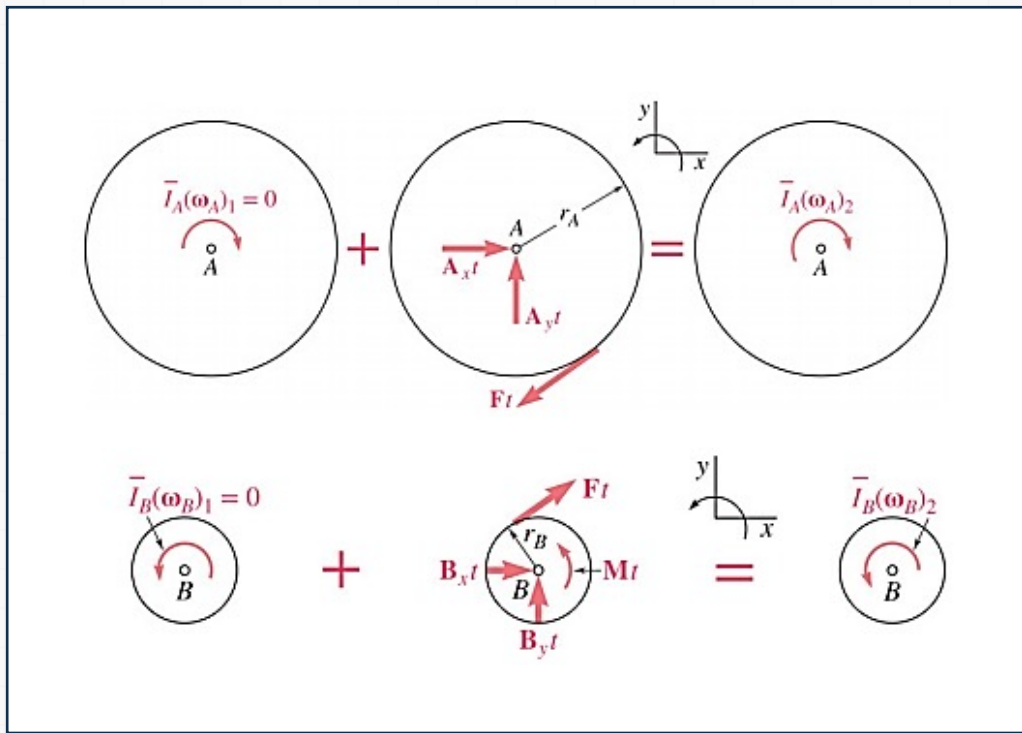
- Considering each gear separately, apply the method of impulse and momentum.
- Solve the angular momentum equations for the two gears simultaneously for the unknown time and tangential force.

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Sample Problem 1

⚙ Modeling and Analysis

- Considering each gear separately, apply the method of impulse and momentum.



+ ∫ moments about A

$$0 - Ftr_A = -I_A(\omega_A)_2$$

$$Ft(0.250) = (0.400)(25.1)$$

$$Ft = 40.2 \text{ N} \cdot \text{s}$$

+ ∫ moments about B

$$0 + Mt - Ftr_B = I_B(\omega_B)_2$$

$$(6)t - Ft(0.100)$$

$$= (0.0192)(62.8)$$

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Sample Problem 1

⚙ Modeling and Analysis

- Solve the angular momentum equations for the two gears simultaneously for the unknown time and tangential force.

$$t = 0.871s \quad F = 46.2N$$

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Sample Problem 1

⚙️ Reflect and Think

This is the same answer obtained in Sample Prob. 2 in Week 14 by the method of work and energy, as you would expect.

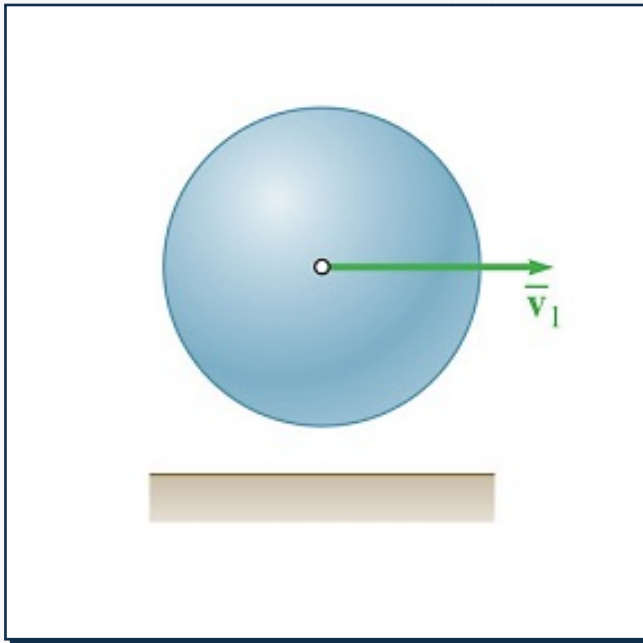
The difference is that in Sample Prob. 2 in Week 14, you were asked to find the number of revolutions, and in this problem, you were asked to find the time.

What you are asked to find will often determine the best approach to use when solving a problem.

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Sample Problem 2

⚙ Determine the time and the linear and angular velocities



- Uniform sphere of mass m and radius r is projected along a rough horizontal surface with a linear velocity $\vec{v}_1$ and no angular velocity.

The coefficient of kinetic friction is μk .

- Determine a) and b)

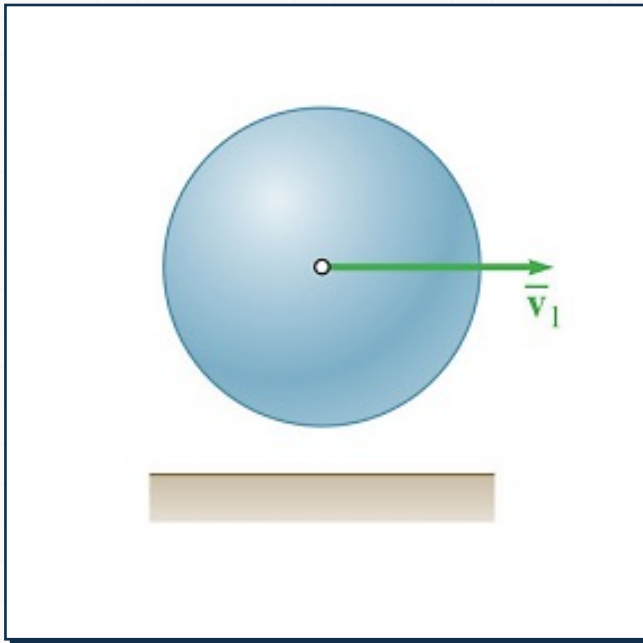
a) the time t_2 at which the sphere will start rolling without sliding

b) the linear and angular velocities of the sphere at time t_2 .

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Sample Problem 2

⚙️ **Determine the time and the linear and angular velocities**



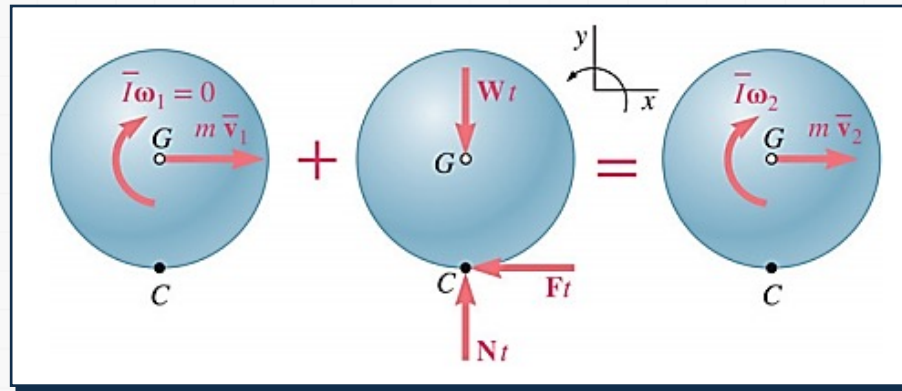
- **Strategy**

- Apply **principle of impulse and momentum** to find variation of linear and angular velocities with time.
 - Relate the linear and angular velocities when the sphere stops sliding by noting that the velocity of the point of contact is zero at that instant.
-
- Substitute for the linear and angular velocities and solve for the time at which sliding stops.
 - Evaluate the linear and angular velocities at that instant.

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Sample Problem 2

Modeling and Analysis



$$\text{Sys Momenta}_1 + \text{Sys Ext Imp}_{1-2} = \text{Sys Momenta}_2$$

+ $\uparrow$ y components

$$N_t - W_t = 0 \quad N = W = mg$$

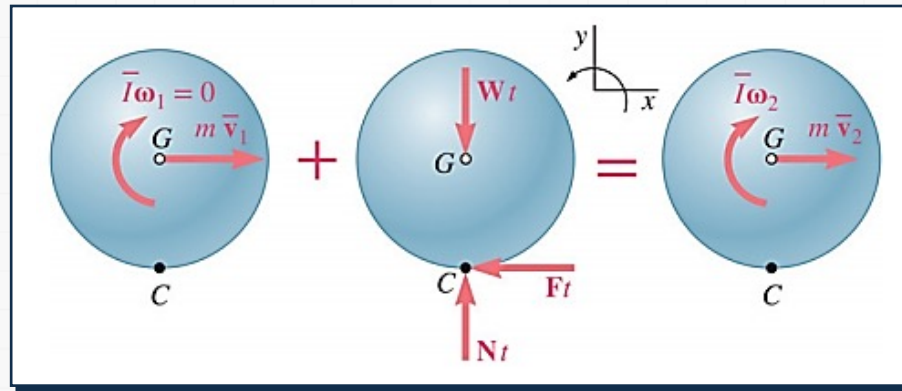
+ $\rightarrow$ x components

$$\begin{aligned} m\bar{v}_1 - Ft &= m\bar{v}_2 & \bar{v}_2 &= \bar{v}_1 - \mu_k gt \\ m\bar{v}_1 - \mu_k mgt &= m\bar{v}_2 \end{aligned}$$

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Sample Problem 2

⚙ Modeling and Analysis



+ ↺ moments about G

$$Ftr = \bar{I}\omega_2$$

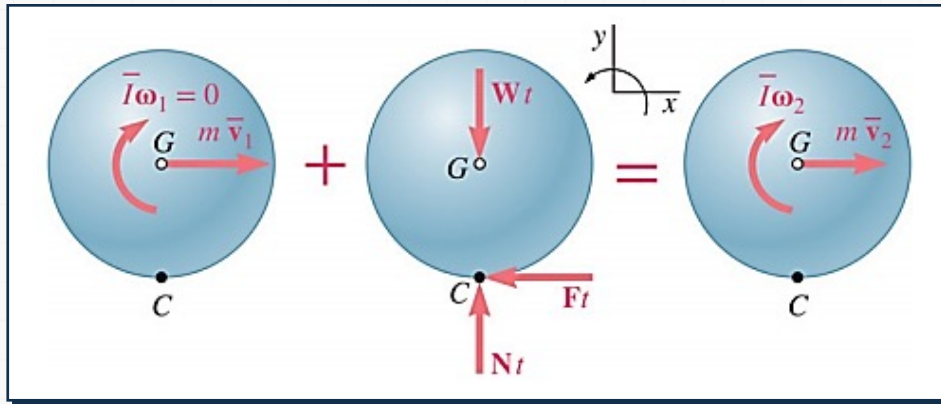
$$(\mu_k mg)tr = \left(\frac{2}{5}mr^2\right)\omega_2$$

$$\omega_2 = \frac{5}{2} \frac{\mu_k g}{r} t$$

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Sample Problem 2

⚙ Modeling and Analysis



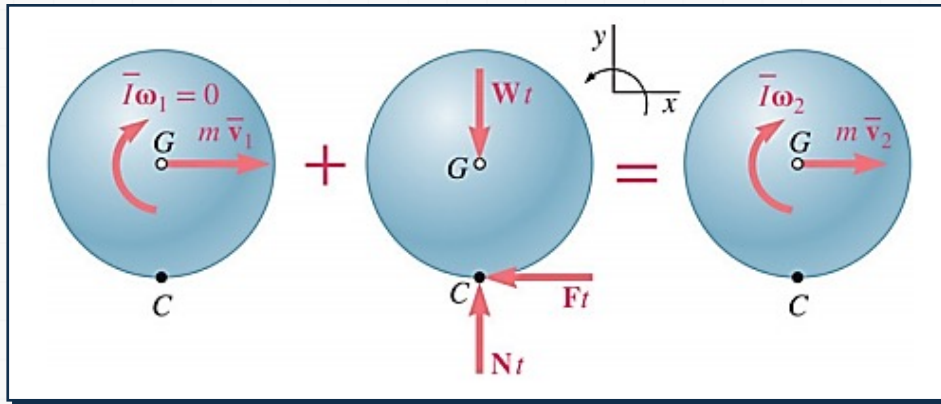
- Apply **principle of impulse and momentum** to find variation of linear and angular velocities with time.

- Relate linear and angular velocities when sphere stops sliding by noting that velocity of point of contact is zero at that instant.

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Sample Problem 2

⚙ Modeling and Analysis



- Substitute for the linear and angular velocities and solve for the time at which sliding stops.

$$\bar{v}_2 = r\omega_2$$

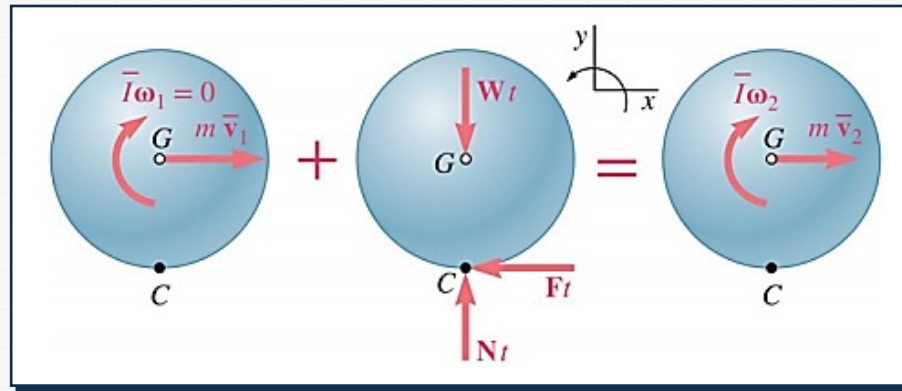
$$\bar{v}_1 - \mu_k g t = r \left(\frac{5}{2} \frac{\mu_k g}{r} t \right)$$

$$t = \frac{2}{7} \frac{\bar{v}_1}{\mu_k g}$$

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Sample Problem 2

⚙ Modeling and Analysis



$$\text{Sys Momenta}_1 + \text{Sys Ext Imp}_{1-2} = \text{Sys Momenta}_2$$

+ $\uparrow$ y components

$$N = W = mg$$

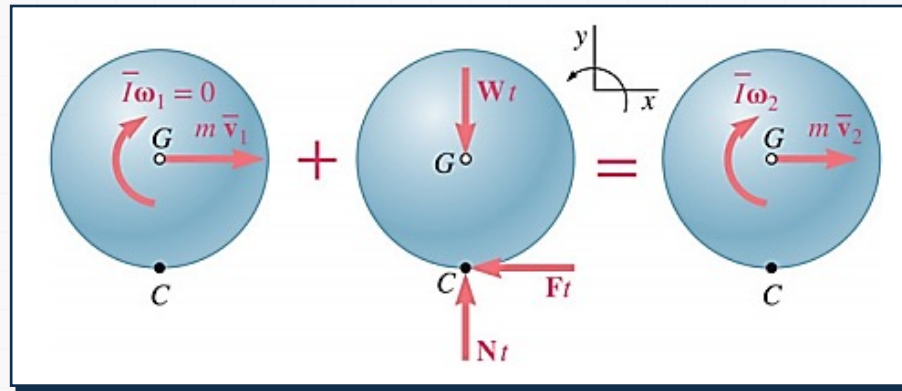
+ $\rightarrow$ x components

$$\bar{v}_2 = \bar{v}_1 - \mu_k g t$$

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Sample Problem 2

⚙ Modeling and Analysis



+ ↺ moments about G

$$\omega_2 = \frac{5 \mu_k g}{2 r} t$$

$$\bar{v}_2 = r \omega_2$$

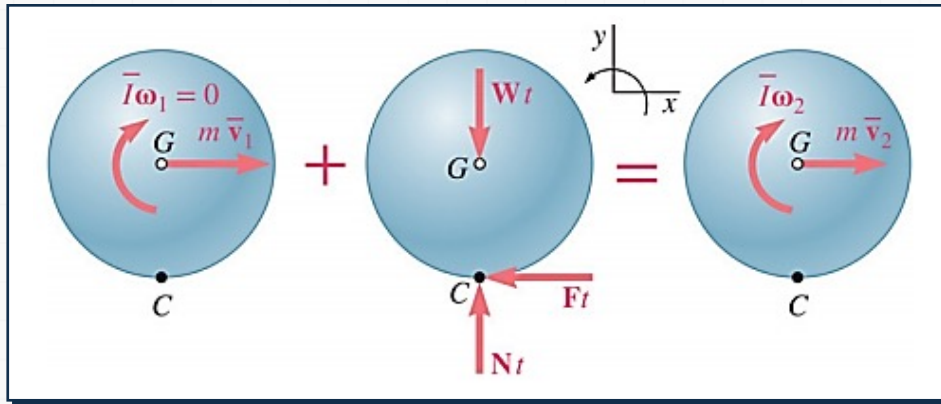
$$\bar{v}_1 - \mu_k g t = r \left(\frac{5 \mu_k g}{2 r} t \right)$$

$$t = \frac{2 \bar{v}_1}{7 \mu_k g}$$

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Sample Problem 2

⚙ Modeling and Analysis



- Evaluate the linear and angular velocities at that instant.

$$\bar{v}_2 = \bar{v}_1 - \mu_k g \left(\frac{2}{7} \frac{\bar{v}_1}{\mu_k g} \right)$$

$$\bar{v}_2 = \frac{5}{7} \bar{v}_1 \rightarrow$$

$$\omega_2 = \frac{5}{2} \frac{\mu_k g}{r} \left(\frac{2}{7} \frac{\bar{v}_1}{\mu_k g} \right)$$

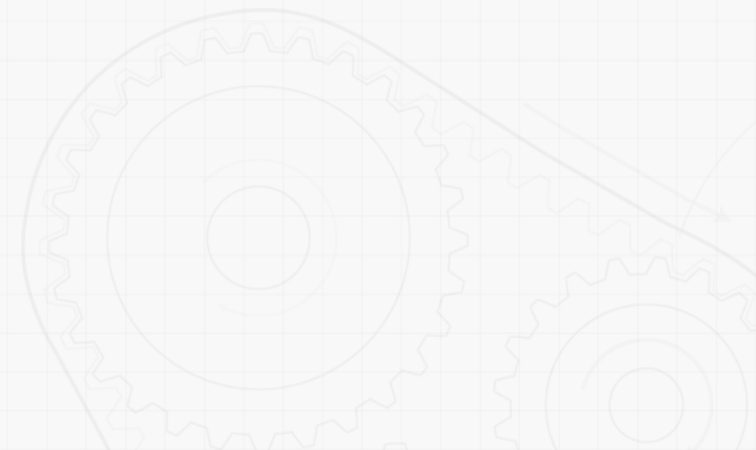
$$\omega_2 = \frac{5}{7} \frac{\bar{v}_1}{r} \curvearrowright$$

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Sample Problem 2

⚙️ Reflect and Think

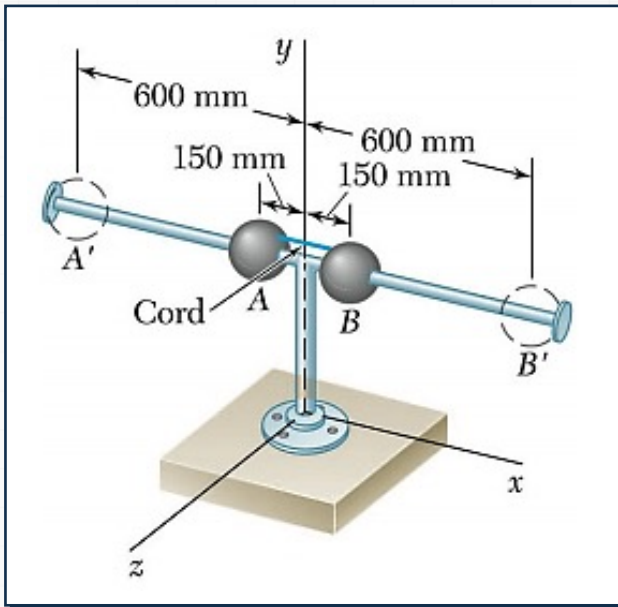
This is the same answer obtained in Sample Prob. 5 in Week 13 by first dealing directly with force and acceleration and then applying kinematic relationships.



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Sample Problem 3

⚙ Determine angular velocity and the energy lost



- Two solid spheres (radius = 100 mm, $m = 1$ kg) are mounted on a spinning horizontal rod ($\bar{I}_R = 0.4$ kg·m², $\omega = 6$ rad/sec) as shown. The balls are held together by a string which is suddenly cut.

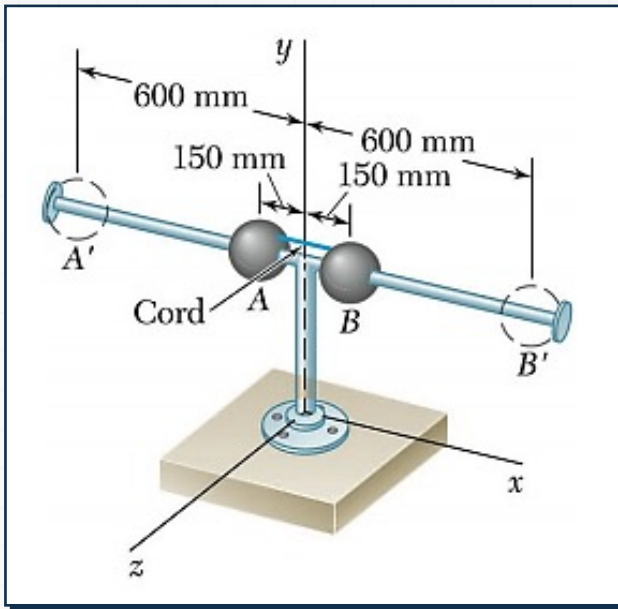
- Determine a) and b)

- a) **angular velocity of the rod** after the balls have moved to A' and B'
- b) **the energy lost** due to the plastic impact of the spheres and stops.

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Sample Problem 3

⚙ Determine angular velocity and the energy lost



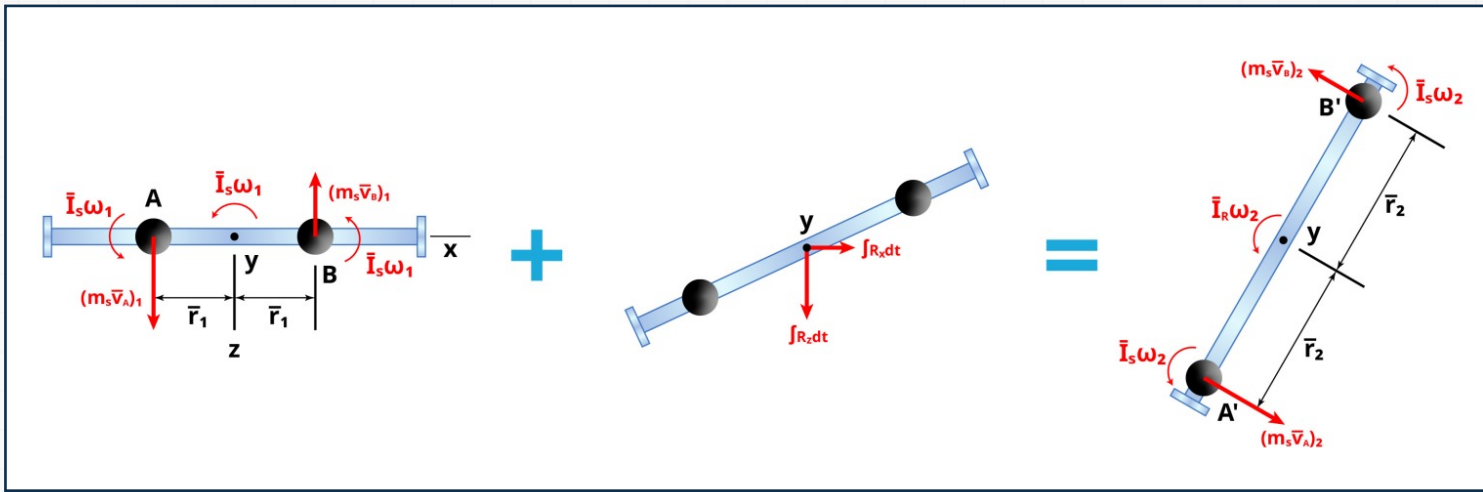
• Strategy

- Observing that none of the external forces produce a moment about the y axis, the angular momentum is conserved.
- Equate the initial and final angular momenta.
Solve for the final angular velocity.
- The energy lost due to the plastic impact is **equal to the change in kinetic energy** of the system.

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Sample Problem 3

⚙ Modeling and Analysis

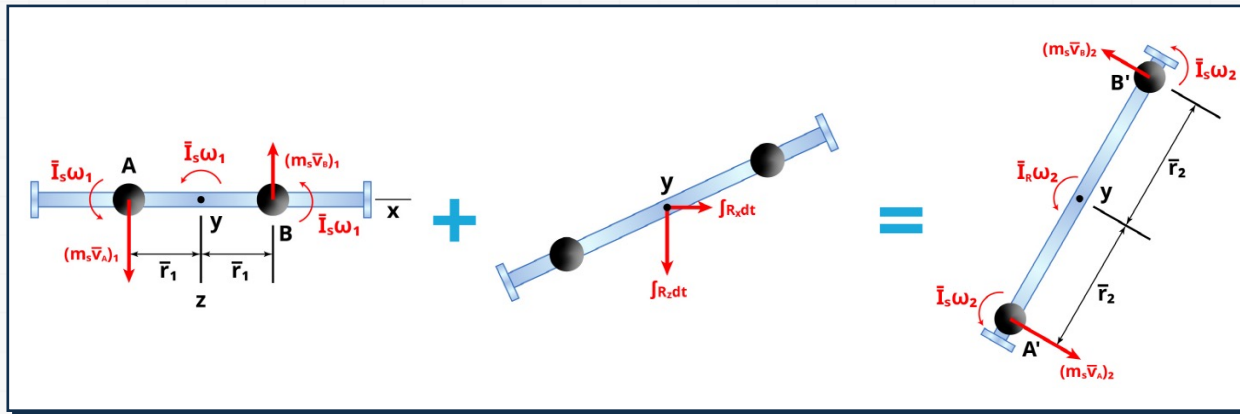


- Observing that none of the external forces produce a moment about the y axis, the angular momentum is conserved.
- Equate the initial and final angular momenta. Solve for the final angular velocity.

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Sample Problem 3

⚙️ Modeling and Analysis



$$\text{Sys Momenta}_1 + \text{Sys Ext Imp}_{1-2} = \text{Sys Momenta}_2$$

$$2[(m_S \bar{r}_1 \omega_1) \bar{r}_1 + \bar{I}_S \omega_1] + \bar{I}_R \omega_1 = 2[(m_S \bar{r}_2 \omega_2) \bar{r}_2 + \bar{I}_S \omega_2] + \bar{I}_R \omega_2$$

$$\omega_2 = \omega_1 \frac{m_S \bar{r}_1^2 + \bar{I}_S + \bar{I}_R}{m_S \bar{r}_2^2 + \bar{I}_S + \bar{I}_R}$$

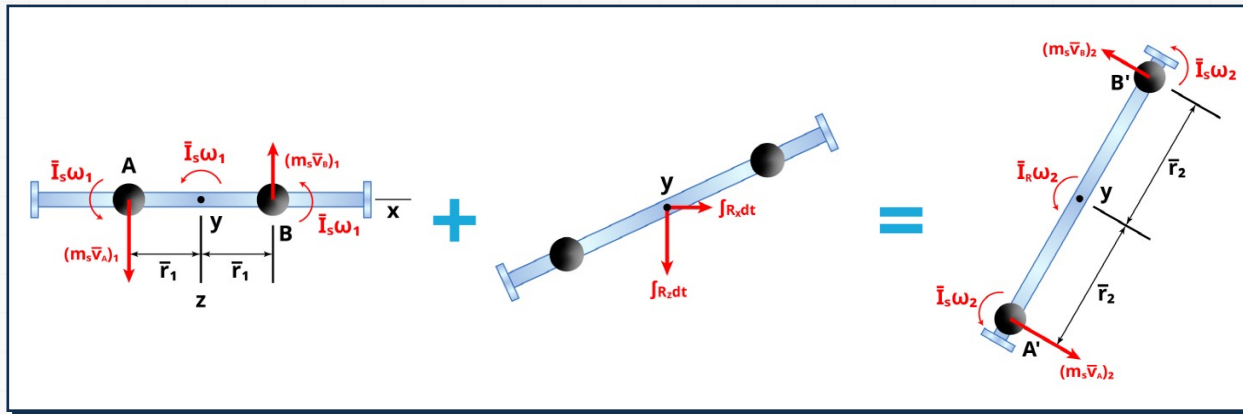
$$\omega_1 = 6 \text{ rad/s}$$

$$\bar{I}_R = 0.4 \text{ kgm}^2$$

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Sample Problem 3

⚙ Modeling and Analysis



$$\bar{I}_S = \frac{2}{5}ma^2 = \frac{2}{5}(1)(0.1)^2 = 0.04 \text{ kgm}^2$$

$$m_S \bar{r}_1^2 = (1)(0.15)^2 = 0.0225 \text{ kgm}^2$$

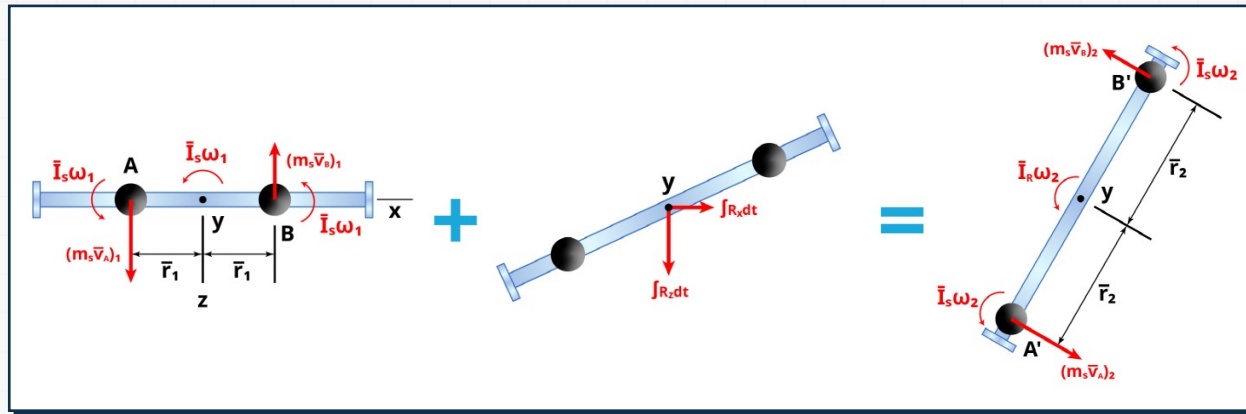
$$m_S \bar{r}_2^2 = (1)(0.6)^2 = 0.36 \text{ kgm}^2$$

$$\omega_2 = 2.41 \text{ rad/s} \curvearrowright$$

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Sample Problem 3

Modeling and Analysis



- The energy lost due to the plastic impact is equal to the change in kinetic energy of the system.

$$\omega_1 = 6 \text{ rad/s}$$

$$\bar{I}_R = 0.4 \text{ kgm}^2$$

$$m_S \bar{r}_1^2 = 0.0225 \text{ kgm}^2$$

$$\omega_2 = 2.4096 \text{ rad/s}$$

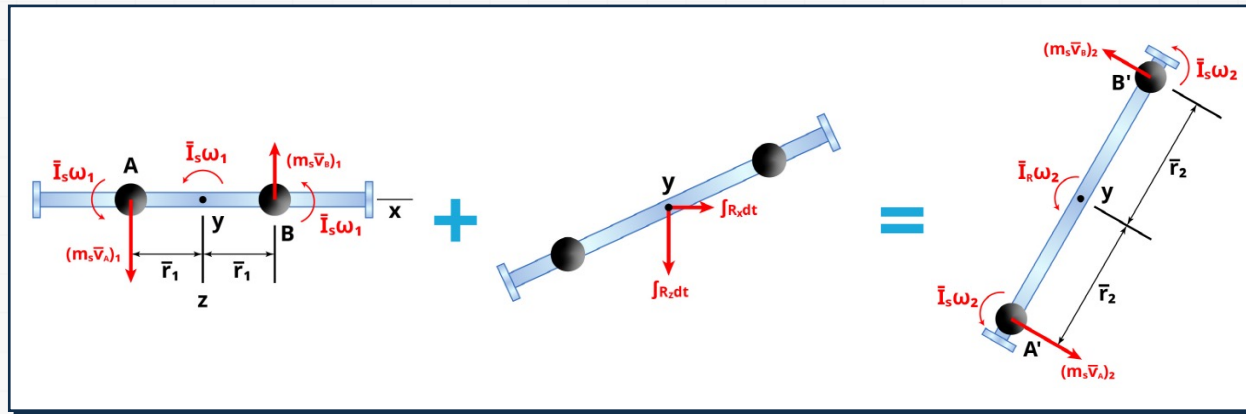
$$\bar{I}_S = 0.004 \text{ kgm}^2$$

$$m_S \bar{r}_2^2 = 0.36 \text{ kgm}^2$$

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Sample Problem 3

⚙ Modeling and Analysis



- The energy lost due to the plastic impact is equal to the change in kinetic energy of the system.

$$T = 2\left(\frac{1}{2}m_S\bar{v}^2 + \frac{1}{2}\bar{I}_S\omega^2\right)\bar{I}_S + \frac{1}{2}\bar{I}_R\omega^2 = \frac{1}{2}(2m_S\bar{r}^2 + 2\bar{I}_S + \bar{I}_R)\omega^2$$

$$T_1 = \frac{1}{2}(0.453)(6)^2 = 8.154J$$

$$T_2 = \frac{1}{2}(1.128)(2.4096)^2 = 3.275J$$

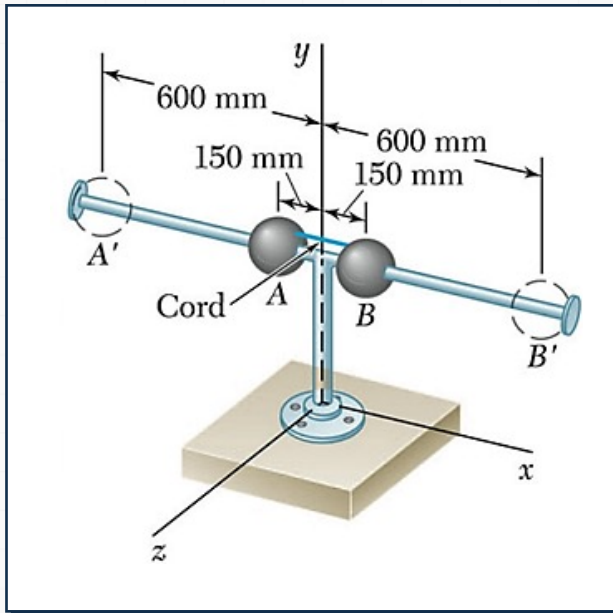
$$\Delta T = T_2 - T_1 = 8.154 - 3.275$$

$$DT = -4.88J$$

15 Impulse and momentum principle in planar motion

Sample Problem 3

⚙️ Reflect and Think



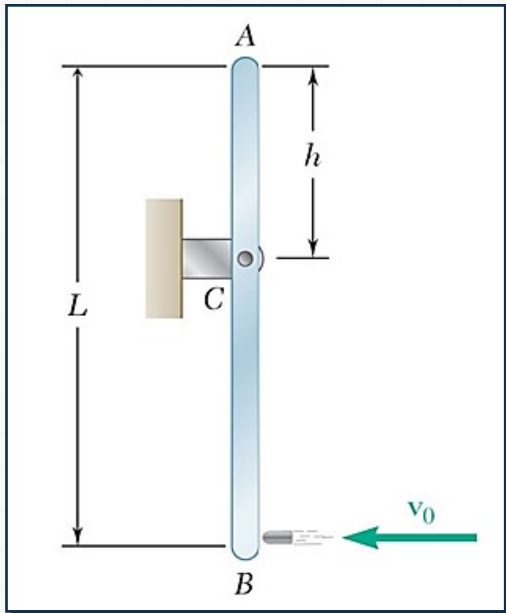
As expected, when the spheres move outward, **the angular velocity of the system decreases.**

This is similar to an ice skater who throws her arms outward to reduce her angular speed.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Determine the angular velocity of the bar



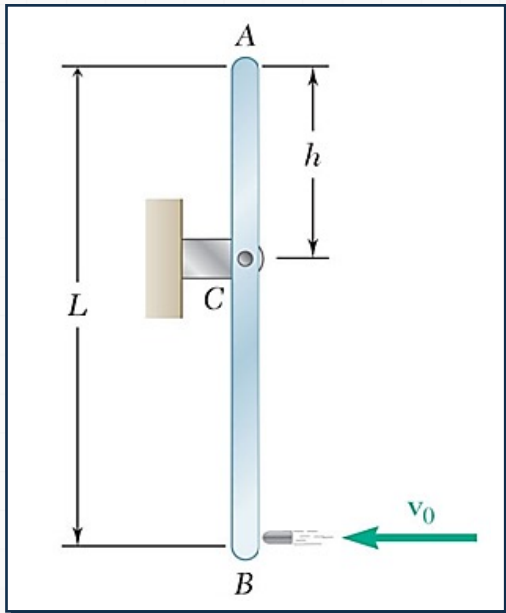
- A projectile weighing 40g is fired with a horizontal velocity of 550 m/s into the lower end of a slender 7.5-kg bar of length $L = 800$ mm.

Knowing that $h = 300$ mm and that the bar is initially at rest, **determine the angular velocity of the bar** when it reaches the horizontal position.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Determine the angular velocity of the bar



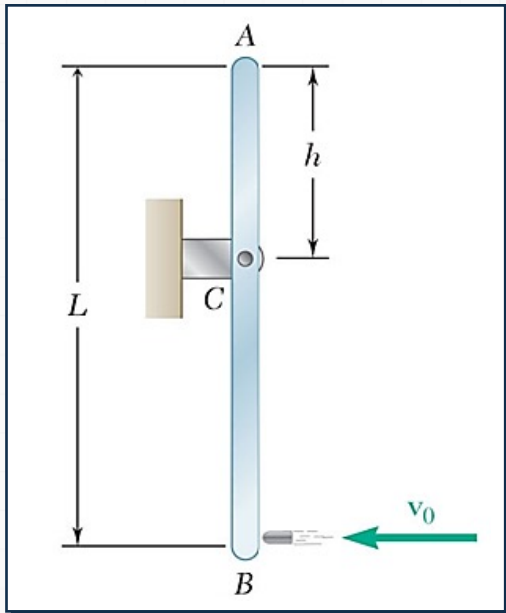
• Strategy

- Consider the projectile and bar as a single system. Apply the **principle of impulse and momentum**.
- The moments about C of the momenta and impulses provide a relation between the final angular velocity of the rod and velocity of the projectile.
- Use the **principle of work-energy** to determine the angle through which the bar swings.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



Given

$$W_o = (0.04)(9.81) = 0.3924N,$$

$$V_o = 550m/s$$

$$W_{AB} = (7.5)(9.81) = 73.575N$$

$$L = 0.8m$$

$$h = 0.3m$$

Find

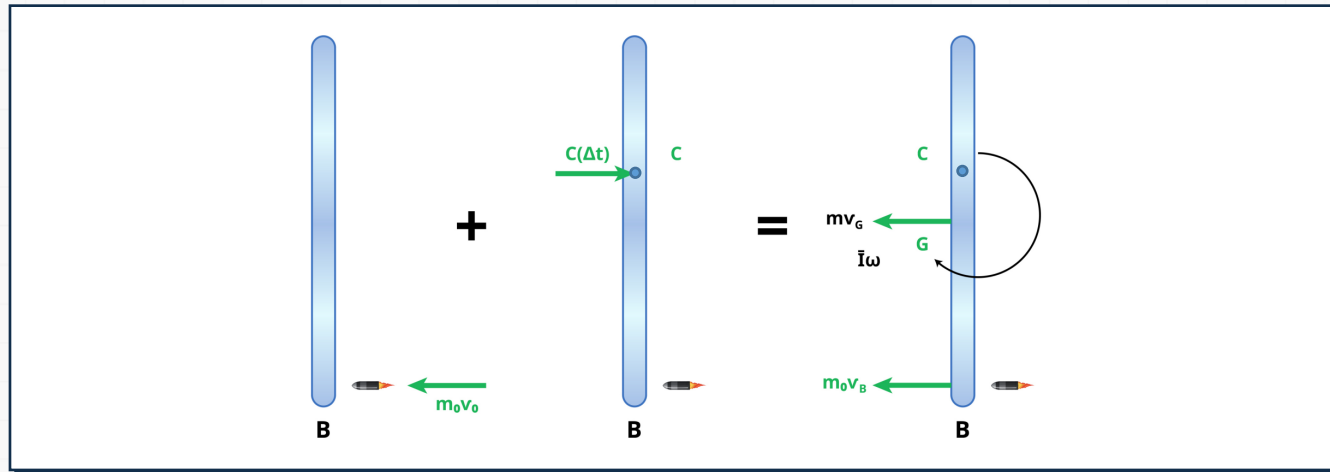
$$\omega_{AB} \text{ when } \theta = 90^\circ$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis

- Draw the impulse momentum diagram



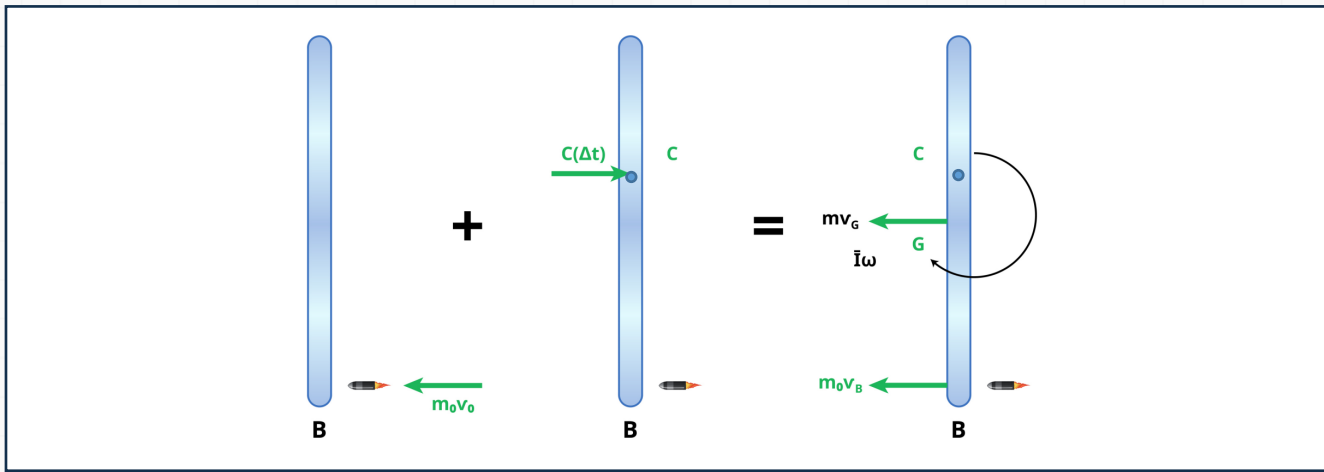
- Apply the angular impulse momentum equation about point C

$$m_0 v_0 (L - h) = m_0 v_B (L - h) + I_C \omega$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Or you could use the relationship:

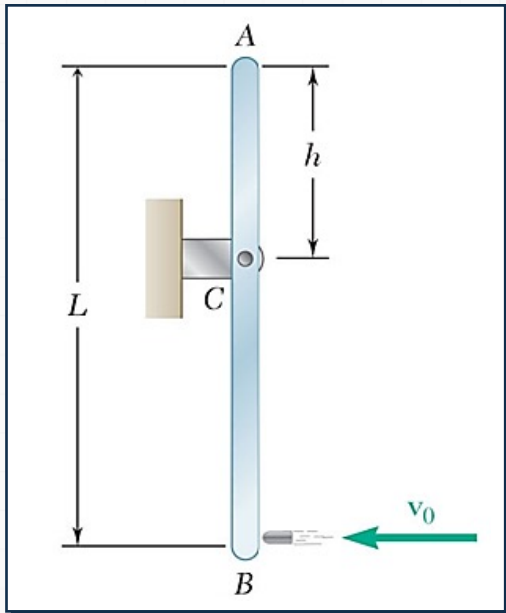
$$m_0 v_0 (L - h) = m_0 v_B (L - h) + m v_0 \left(\frac{L}{2} - h \right) + \bar{I} \omega$$

$$m_0 v_0 (L - h) = m_0 v_B (L - h) + I_C \omega$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Relate v_B and ω (after the impact)

$$v_B = (L - h)\omega$$

- Substitute into equation (1) and solve for ω

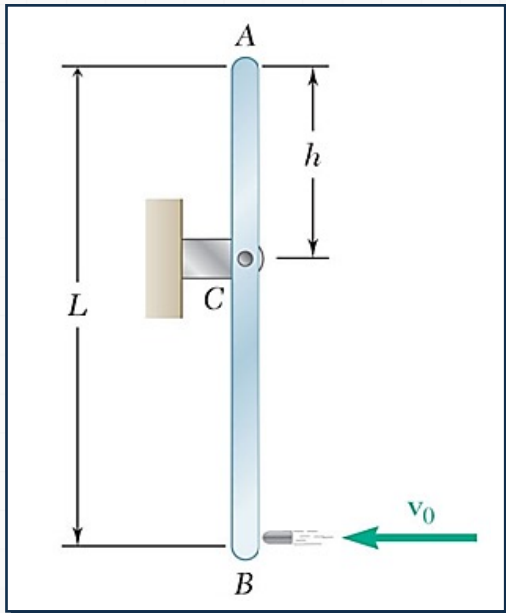
$$m_0 v_0 (L - h) = m_0 (L - h)^2 \omega + I_C \omega$$

$$\omega = \frac{m_0 v_0 (L - h)}{m_0 (L - h)^2 + I_C}$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



- Find I_C

$$L = 0.8m \quad m = 7.5kg$$

$$I_C = \frac{1}{12}mL^2 + md^2 = \frac{1}{12}(7.5)(0.8)^2 + (7.5)(0.1)^2$$

$$I_C = 0.475kg \cdot m^2$$

- Substitute and solve

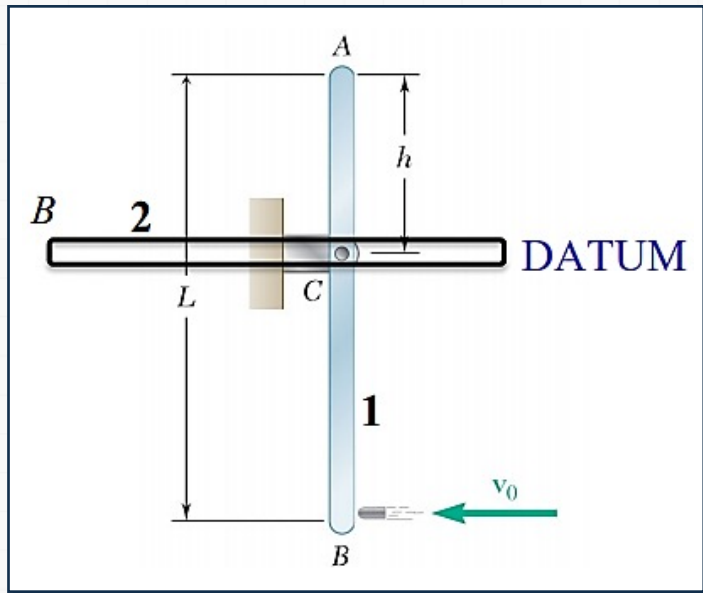
$$\omega = \frac{m_0 v_0 (L - h)}{m_0 (L - h)^2 + I_C} = \frac{(0.04)(550)(0.8 - 0.3)}{(0.04)(0.8 - 0.3)^2 + 0.475}$$

$$\omega_2 = 22.68rad/s$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



- Draw position 1 and 2, set your datum and apply the conservation of energy equation

$$T_2 + V_2 = T_3 + V_3$$

- Find T_2

$$T_2 = \frac{1}{2} I_C \omega_2^2 = \frac{1}{2} (0.475) (22.68)^2 \quad T_2 = 122.166 \text{ J}$$

- Find V_2

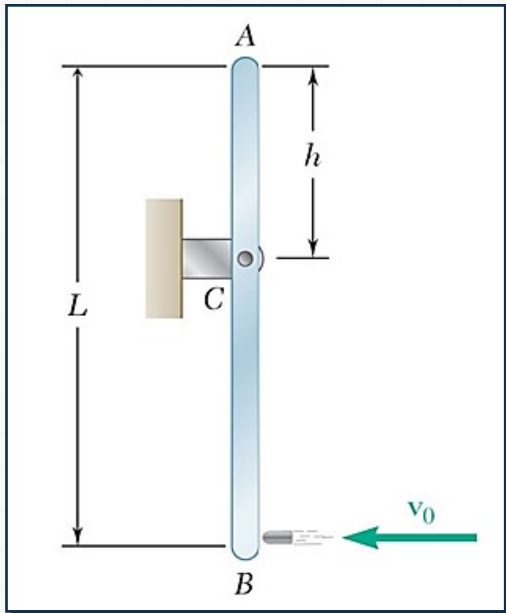
$$V_2 = m_{AB} g y_{AB2} + m_O g y_{O2} = W_{AB} y_{AB2} + W_O y_{O2}$$

$$V_2 = -73.575 \left(\frac{L}{2} - h \right) - 0.3924(L - h) = -73.575(0.1) - (0.3924)(0.5) = -7.5537 \text{ J}$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Solve for ω_3

$$T_3 = \frac{1}{2} I_C \omega_3^2 = T_2 + V_2$$

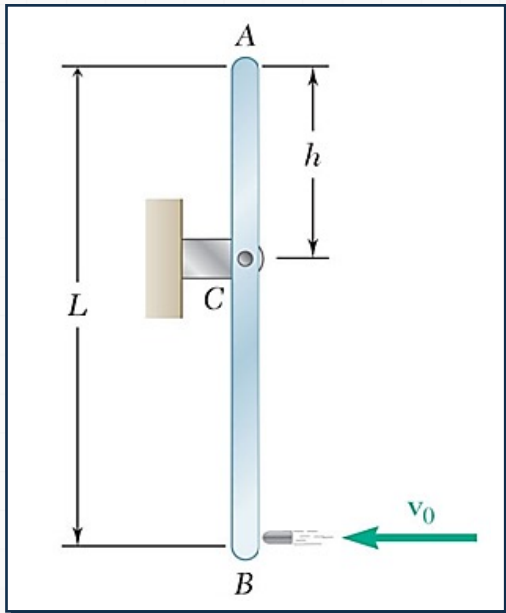
$$\frac{1}{2} (0.475) \omega_3^2 = 122.166 - 7.5537$$

$$\omega_3 = 22.0 \text{ rad/s}$$

15 Impulse and momentum principle in planar motion

Concept Question

⚙ Determine the reaction forces

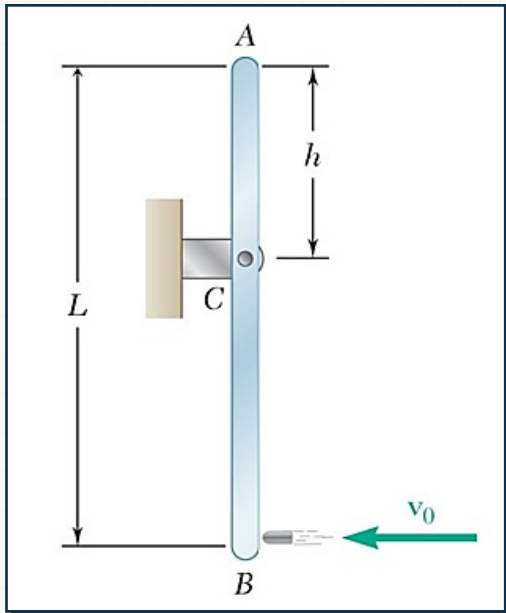


- For the previous problem, how would you determine the reaction forces at C when the bar is horizontal?
 - a) Apply linear-momentum to solve for $C_x\Delta t$ and $C_y\Delta t$.
 - b) Use work-energy to determine the work done by the moment at C .
 - c) Use sum of forces and sum of moments equations when the bar is horizontal.

15 Impulse and momentum principle in planar motion

Concept Question

⚙ Determine the reaction forces

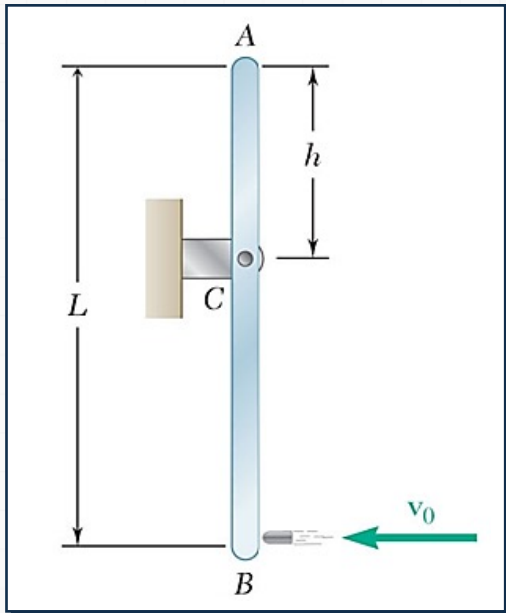


- For the previous problem, how would you determine the reaction forces at C when the bar is horizontal?
 - a) Apply linear-momentum to solve for $C_x\Delta t$ and $C_y\Delta t$.
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 - c) Use sum of forces and sum of moments equations when the bar is horizontal.

15 Impulse and momentum principle in planar motion

Concept Question

⚙ What would happen?

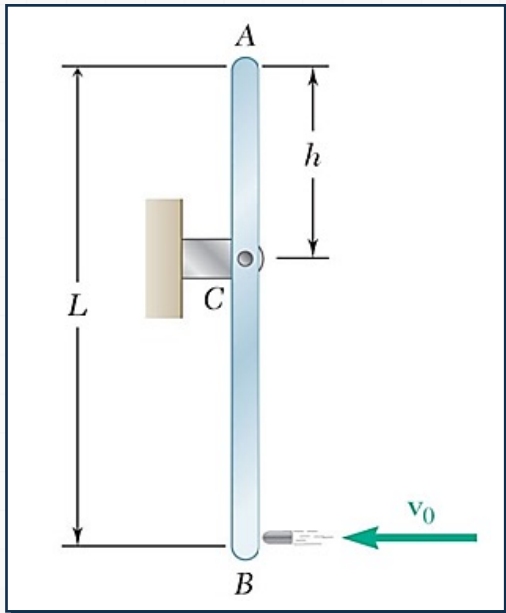


- For the previous problem, what would happen if the coefficient of restitution between the projectile and bar was 1.0 instead of zero?
 - a) The angular velocity after impact would be bigger.
 - b) The angular velocity after impact would be smaller.
 - c) The angular velocity after impact would be the same.
 - d) Not enough information to tell.

15 Impulse and momentum principle in planar motion

Concept Question

⚙️ What would happen?

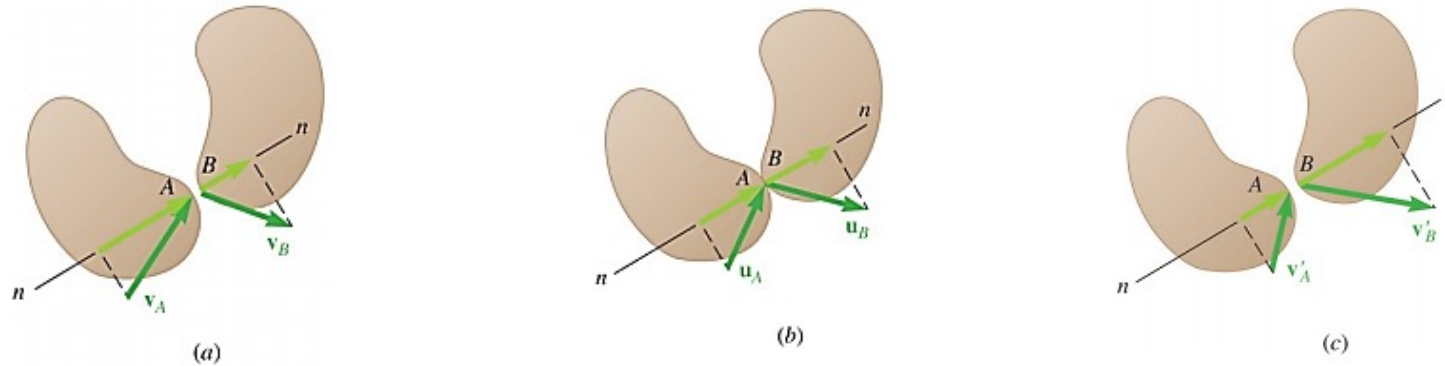


- For the previous problem, what would happen if the coefficient of restitution between the projectile and bar was 1.0 instead of zero?
 - a) The angular velocity after impact would be bigger.
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 - c) The angular velocity after impact would be the same.
 - d) Not enough information to tell.

15 Impulse and momentum principle in planar motion

Eccentric Impact

⚙ Eccentric Impact



$$(\vec{u}_A)_n = (\vec{u}_B)_n$$

Period of deformation

$$\text{Impulse} = \int \vec{R} dt$$

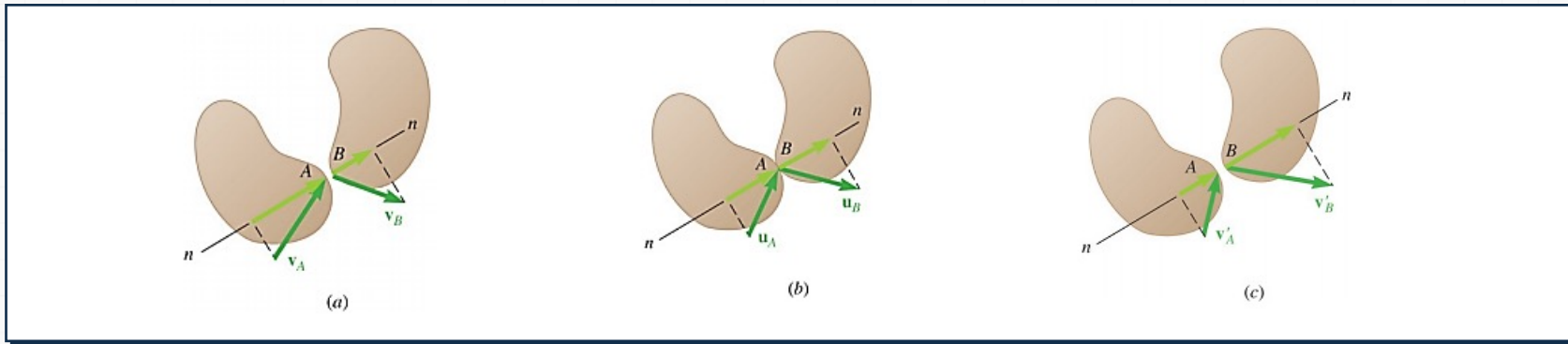
Period of restitution

$$\text{Impulse} = \int \vec{P} dt$$

15 Impulse and momentum principle in planar motion

Eccentric Impact

⚙ Eccentric Impact



- Principle of impulse and momentum is supplemented by

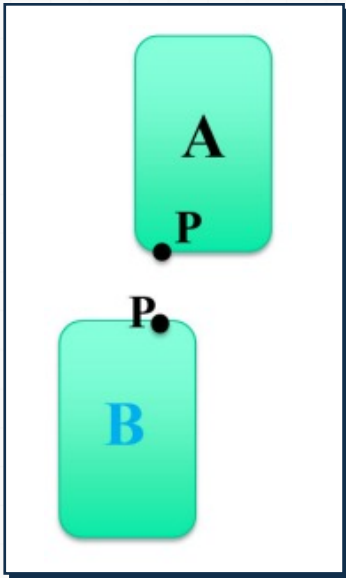
$$e = \text{coefficient of restitution} = \frac{\int \vec{R} dt}{\int \vec{P} dt} = \frac{(v'_B)_n - (v'_A)_n}{(v_A)_n - (v_B)_n}$$

These velocities are for the points of impact.

15 Impulse and momentum principle in planar motion

Concept Question

⚙ Which of the following can you use?

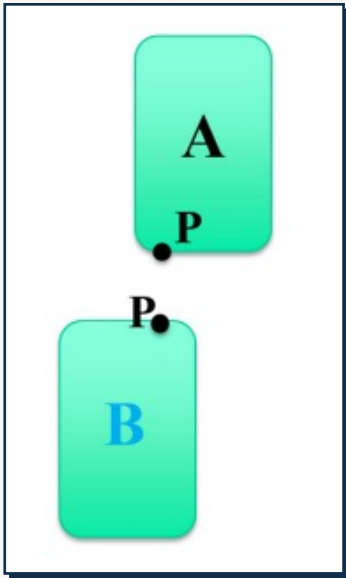


- The cars collide, hitting at point P as shown. Which of the following can you use to help analyze the collision?
 - a) The linear momentum of car A is conserved.
 - b) The linear momentum of the combined two cars is conserved.
 - c) The total kinetic energy before the impact equals the total kinetic energy after the impact.
 - d) The angular momentum about the CG of car B is conserved.

15 Impulse and momentum principle in planar motion

Concept Question

⚙ Which of the following can you use?

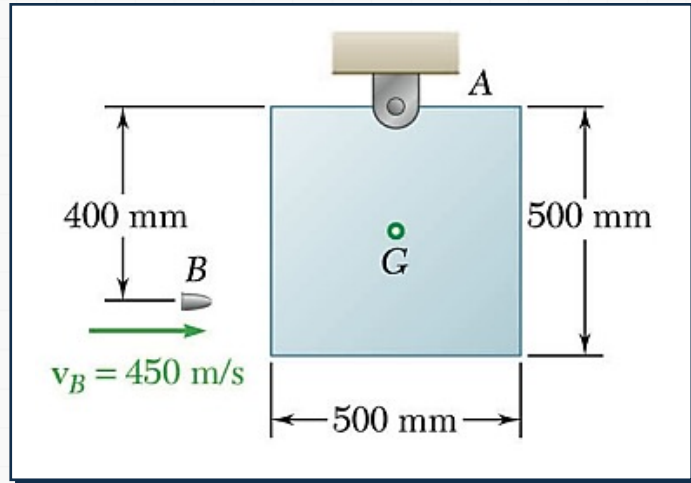


- The cars collide, hitting at point P as shown. Which of the following can you use to help analyze the collision?
 - a) The linear momentum of car A is conserved.
 - b) The linear momentum of the combined two cars is conserved.
 - c) The total kinetic energy before the impact equals the total kinetic energy after the impact.
 - d) The angular momentum about the CG of car B is conserved.

15 Impulse and momentum principle in planar motion

Sample Problem 4

⚙️ Determine the angular velocity and the impulsive reaction



- A 25-g bullet is fired into the side of a 10-kg square panel which is initially at rest.

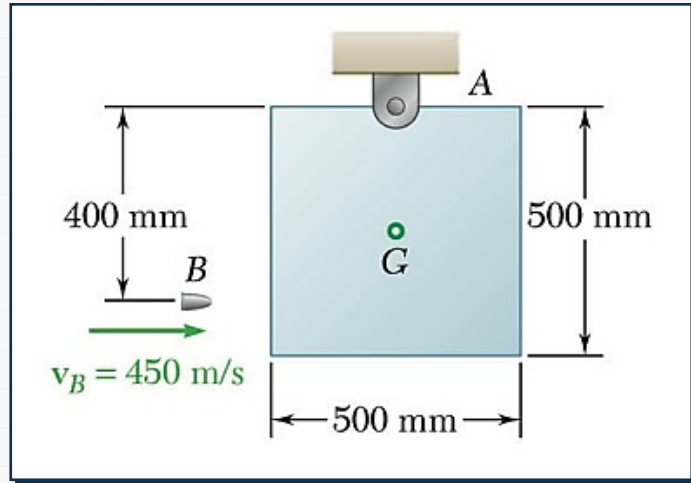
- Determine a) and b)

- the angular velocity of the panel immediately after the bullet becomes embedded
- the impulsive reaction at A, assuming that the bullet becomes embedded in 0.0006 s.

15 Impulse and momentum principle in planar motion

Sample Problem 4

⚙️ Determine the angular velocity and the impulsive reaction



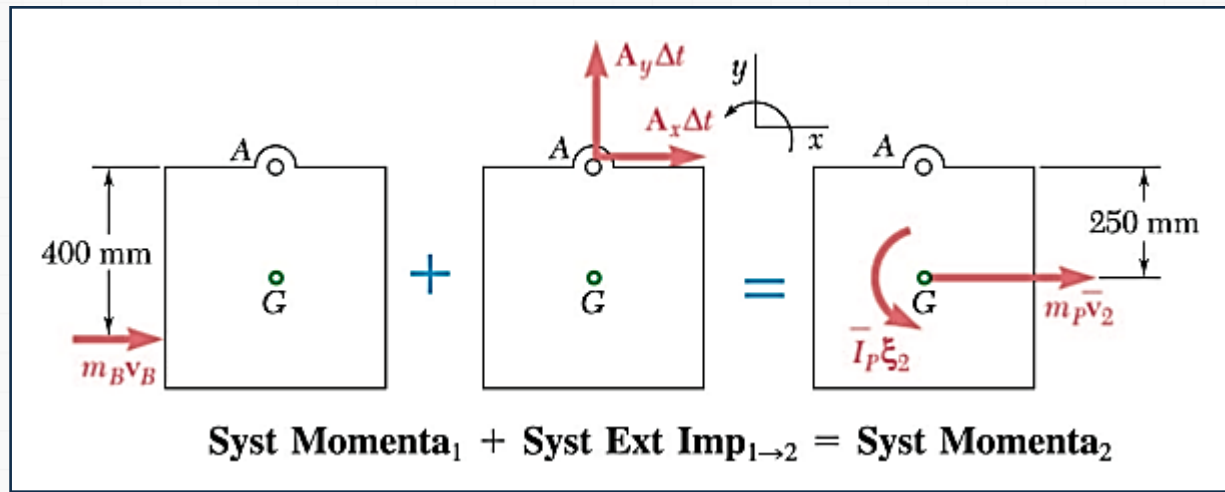
• Strategy

- Consider a system consisting of the bullet and panel. Apply the principle of impulse and momentum.
- The final angular velocity is found from the moments of the momenta and impulses about A.
- The reaction at A is found from the horizontal and vertical momenta and impulses.

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis

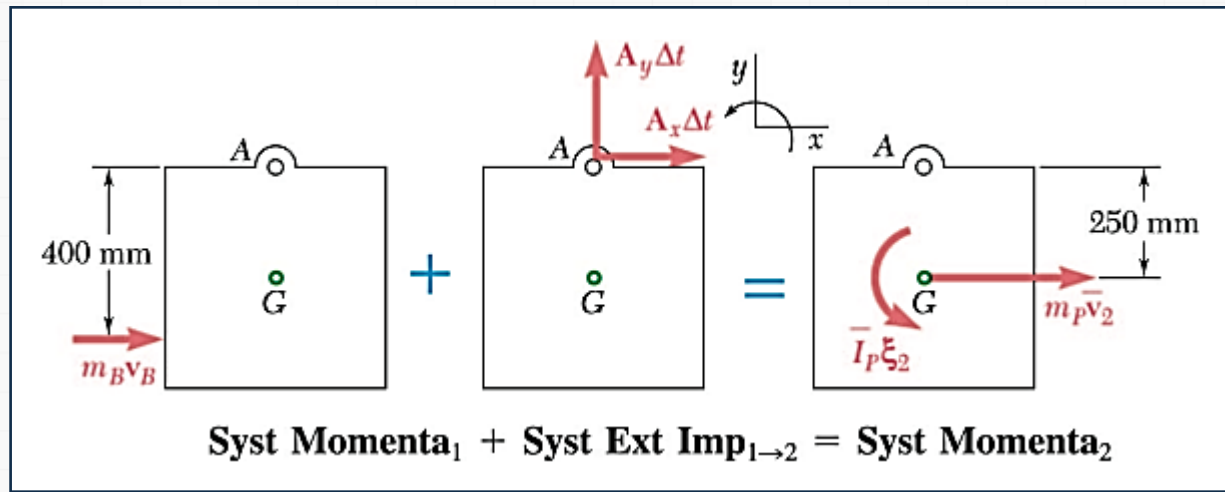


- Consider a system consisting of the bullet and panel. Apply the principle of impulse and momentum.
- The final angular velocity is found from the moments of the momenta and impulses about A.

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis



+ ↺ moments about A

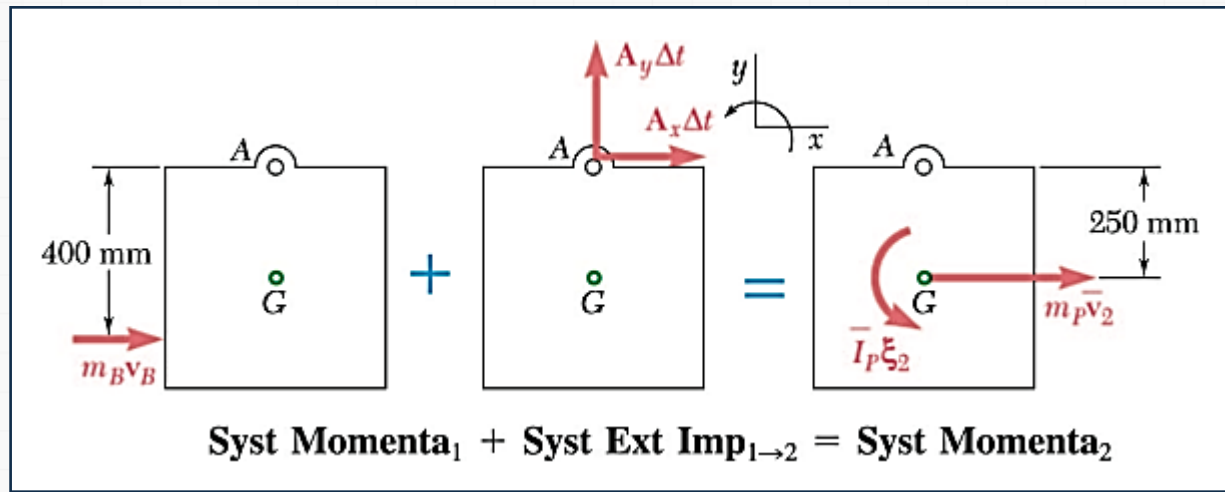
$$m_B v_B (0.4) + 0 = m_P \bar{v}_2 (0.25) + \bar{I}_P \omega_2$$

$$\bar{v}_2 = (0.25) \omega_2 \quad \bar{I}_P = \frac{1}{6} m_P b^2 = \frac{1}{6} (10) (0.5)^2 = 0.417 \text{ kgm}^2$$

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis



$$(0.025)(450)(0.4) = (10)(0.25\omega_2)(0.25) + 0.417\omega_2$$

$$\omega_2 = 4.32 \text{ rad/s}$$

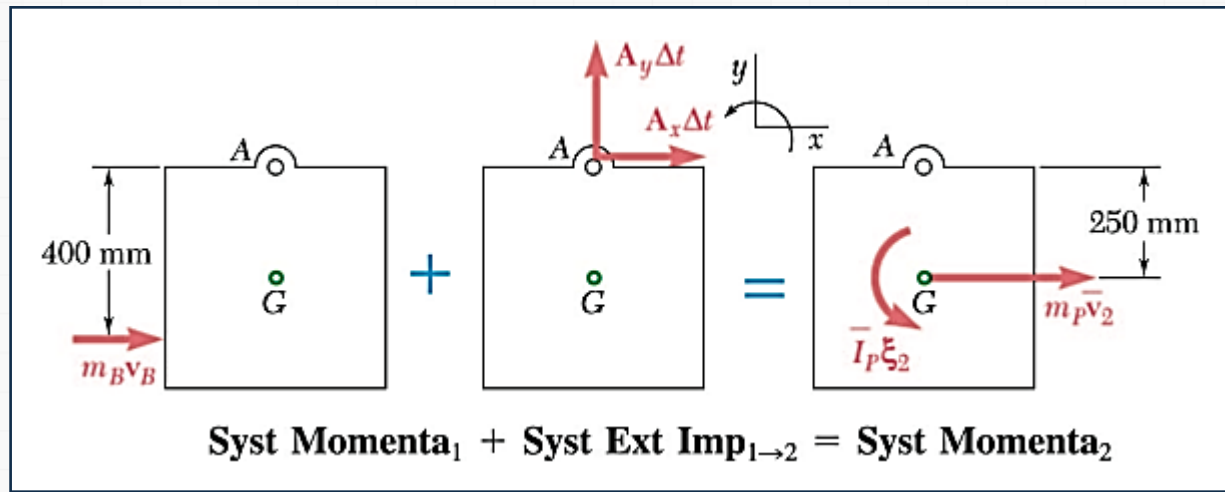
$$\bar{v}_2 = (0.25)\omega_2 = 1.08 \text{ m/s}$$

$$\omega_2 = 4.32 \text{ rad/s } \curvearrowright$$

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis



- The reactions at A are found from the horizontal and vertical momenta and impulses.

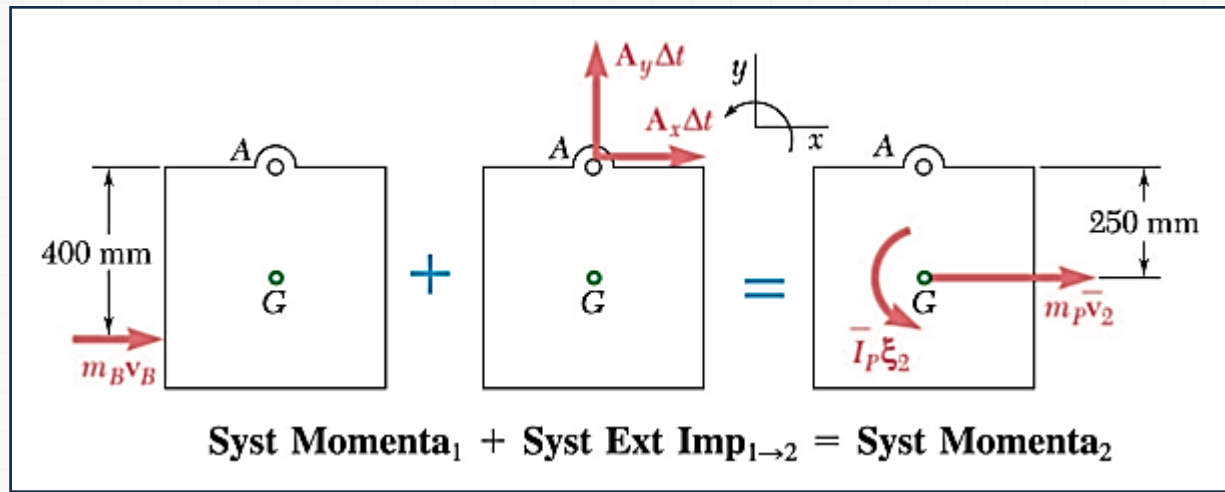
$$\omega_2 = 4.32 \text{ rad/s}$$

$$\bar{v}_2 = (0.25)\omega_2 = 1.08 \text{ m/s}$$

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis



$\rightarrow$ x components

$$m_B v_B + A_x \Delta t = m_p \bar{v}_2 \quad (0.025)(450) + A_x(0.0006) = (10)(1.08)$$

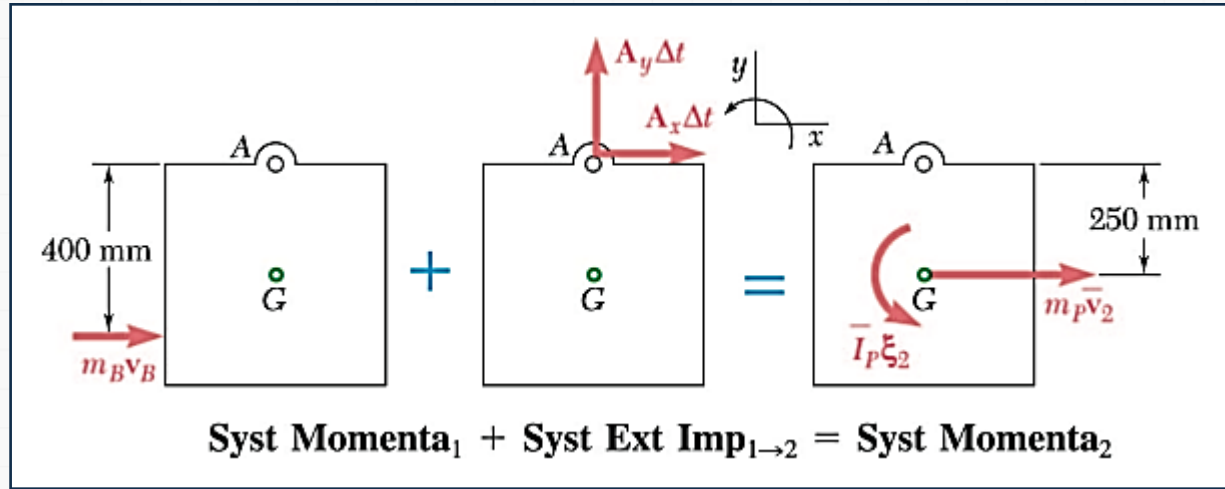
$$A_x = -750 \text{ N}$$

$$A_x = 750 \text{ N} \leftarrow$$

15 Impulse and momentum principle in planar motion

Sample Problem 4

Modeling and Analysis



+↑ y components

$$0 + A_y \Delta t = 0$$

$$A_y = 0$$

15 Impulse and momentum principle in planar motion

Sample Problem 4

⚙️ Reflect and Think

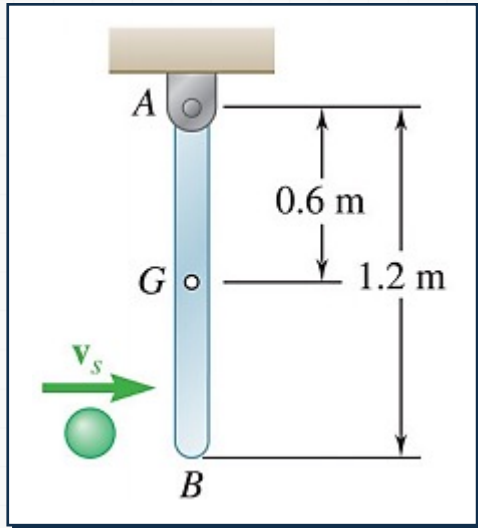
The speed of the bullet is in the range of a modern high-performance rifle.

Notice that the reaction at A is over 5000 times the weight of the bullet and over 10 times the weight of the plate.

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙️ Determine the angular velocity and the velocity of the sphere



- A 2-kg sphere with an initial velocity of 5 m/s strikes the lower end of an 8-kg rod AB .

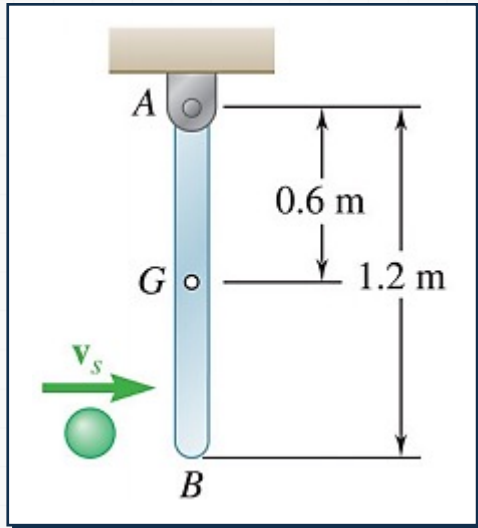
The rod is hinged at A and initially at rest. The coefficient of restitution between the rod and sphere is 0.8.

- Determine the angular velocity of the rod and the velocity of the sphere immediately after impact.

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙️ Determine the angular velocity and the velocity of the sphere



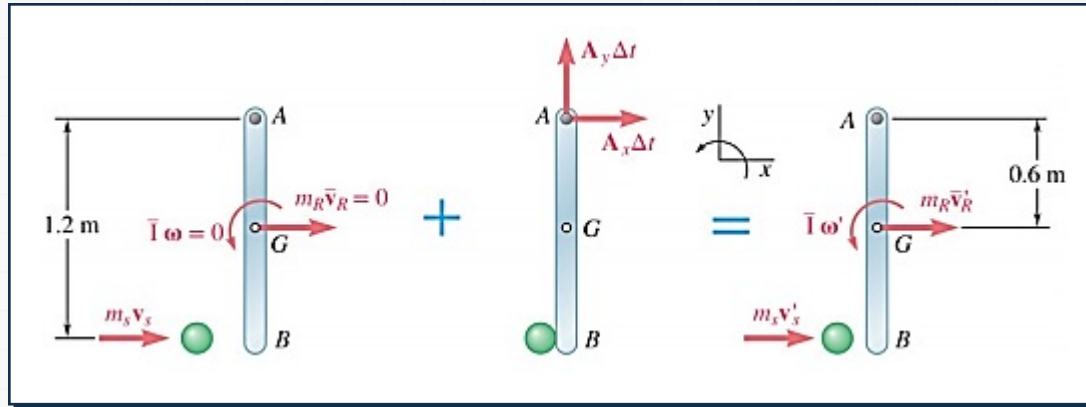
• Strategy

- Consider the sphere and rod as a single system. Apply **the principle of impulse and momentum**.
- The moments about A of the momenta and impulses provide a relation between the final angular velocity of the rod and velocity of the sphere.
- The definition of the coefficient of restitution **provides a second relationship** between the final angular velocity of the rod and velocity of the sphere.
- Solve the **two relations simultaneously** for the angular velocity of the rod and velocity of the sphere.

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis



+ ↺ moments about A

$$m_s v_s (1.2) = m_s v'_s (1.2) + m_R \bar{v}'_R (0.6) + \bar{I} \omega'$$

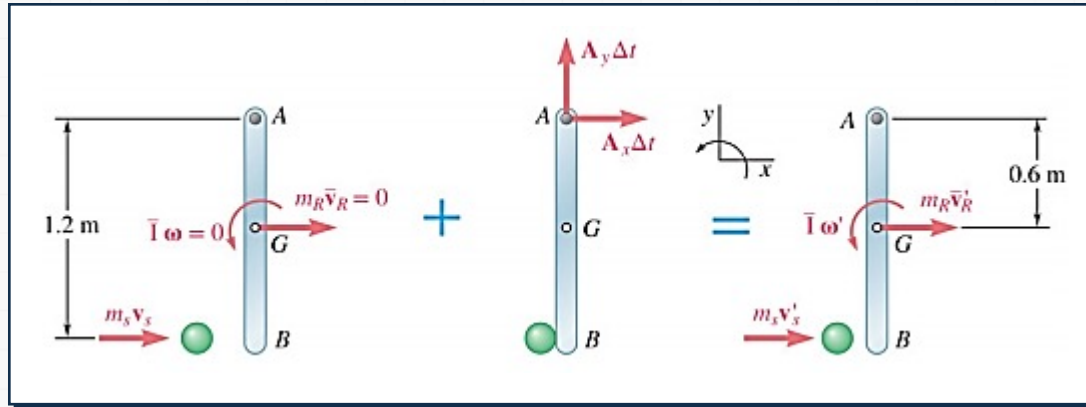
$$\bar{v}'_R = \bar{r} \omega' = (0.6) \omega'$$

$$\bar{I} = \frac{1}{12} m L^2 = \frac{1}{12} (8) (1.2)^2 = 0.96 \text{ kg} \cdot \text{m}^2$$

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis



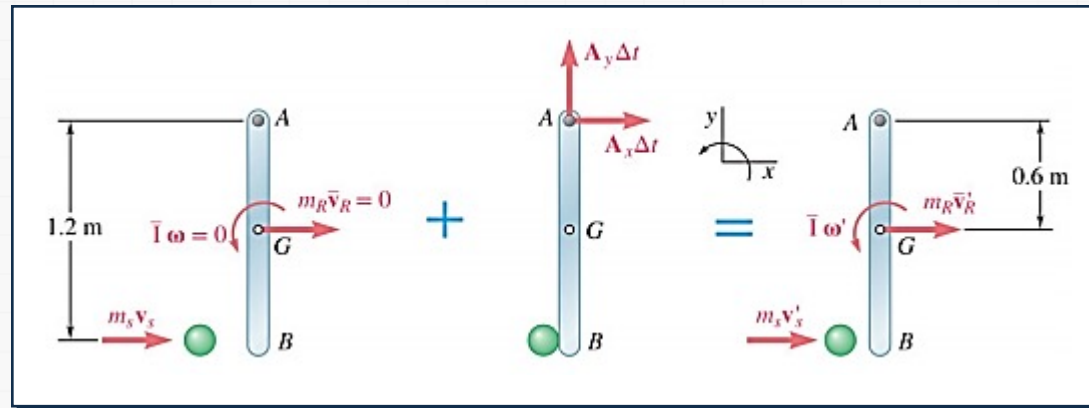
$$(2)(5)(1.2) = (2)v'_s(1.2) + (8)(0.6)\omega'(0.6) + (0.96)\omega'$$

$$12 = 2.4v'_s + 3.84\omega'$$

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis



± Relative velocities

$$v'_B - v'_S = e(v_B - v_S)$$

$$(1.2)\omega' - v'_S = 0.8(5)$$

Solving,

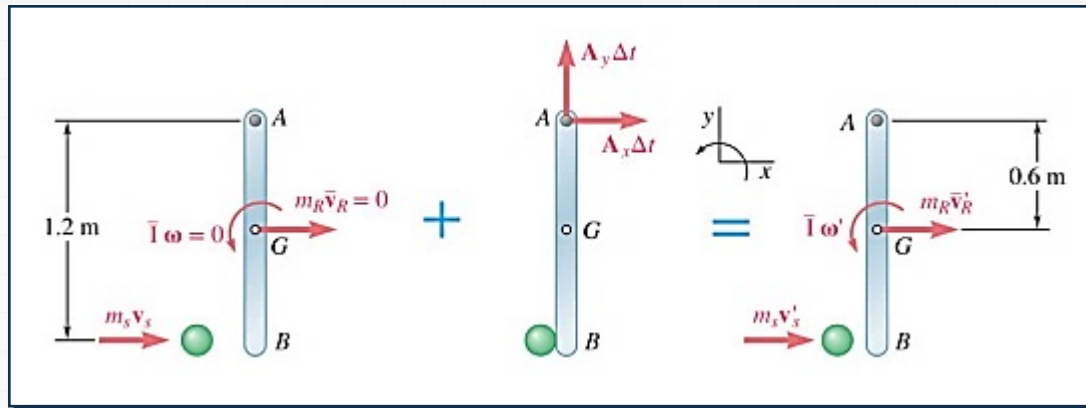
$$\omega' = 3.21 \text{ rad/s} \quad v'_S = -0.143 \text{ m/s}$$

$$\omega' = 3.21 \text{ rad/s} \cup \quad v'_S = 0.143 \text{ m/s} \leftarrow$$

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis

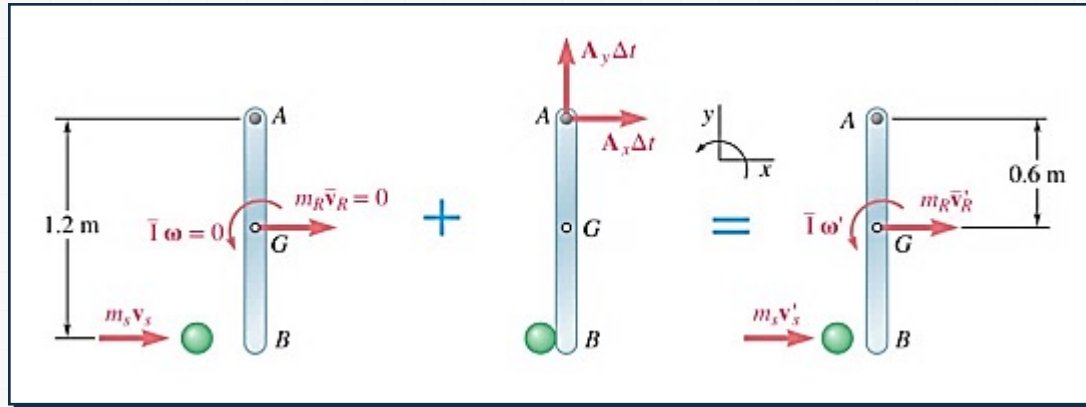


- Consider the sphere and rod as a single system. Apply **the principle of impulse and momentum**.
- The moments about A of the momenta and impulses provide a relation between the final angular velocity of the rod and velocity of the rod.

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙ Modeling and Analysis



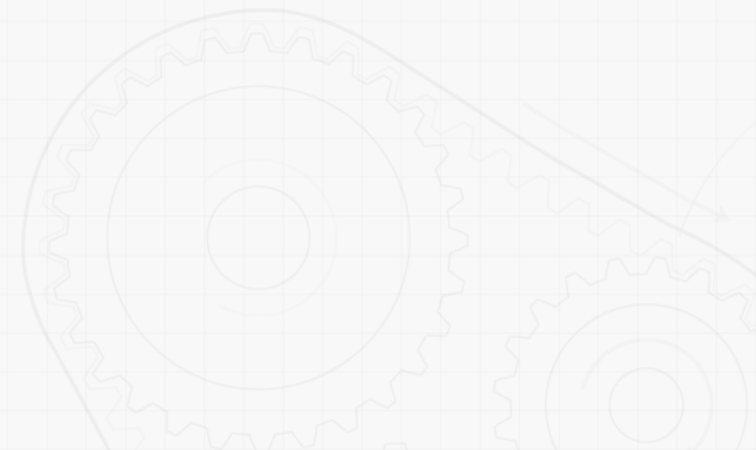
- The definition of the coefficient of restitution provides a second relationship between the final angular velocity of the rod and velocity of the sphere.
- Solve the two relations simultaneously for the angular velocity of the rod and velocity of the sphere.

15 Impulse and momentum principle in planar motion

Sample Problem 5

⚙️ Reflect and Think

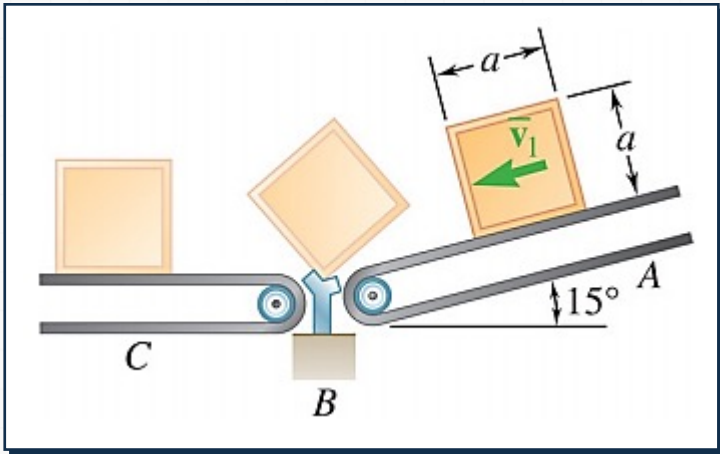
The negative value for the velocity of the sphere after impact means that **it bounces back to the left**. Given the masses of the sphere and the rod, this seems reasonable.



15 Impulse and momentum principle in planar motion

Sample Problem 6

⚙ Derive an expression for the minimum velocity



- A square package of mass m moves down conveyor belt A with constant velocity.

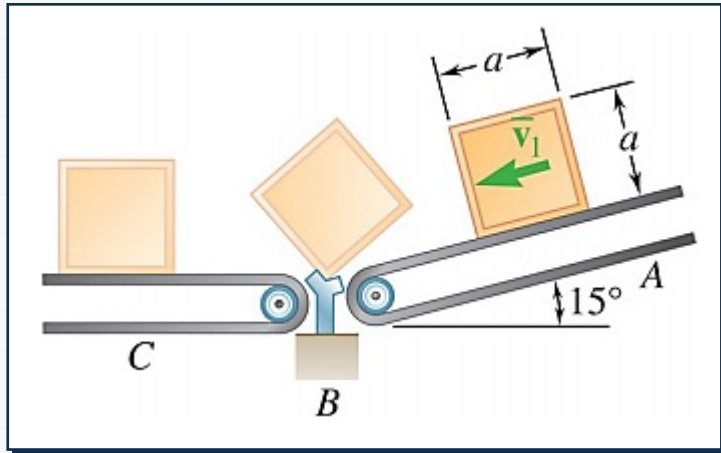
At the end of the conveyor, the corner of the package strikes a rigid support at B . The **impact is perfectly plastic**.

- Derive an **expression for the minimum velocity of conveyor belt A** for which the package will rotate about B and reach conveyor belt C .

15 Impulse and momentum principle in planar motion

Sample Problem 6

⚙️ Derive an expression for the minimum velocity



• Strategy

- Apply the principle of impulse and momentum to relate the velocity of the package on conveyor belt A before the impact at B to the angular velocity about B after impact.
- Apply the principle of conservation of energy to determine the minimum initial angular velocity such that the mass center of the package will reach a position directly above B.
- Relate the required angular velocity to the velocity of conveyor belt A.

15 Impulse and momentum principle in planar motion

Sample Problem 6

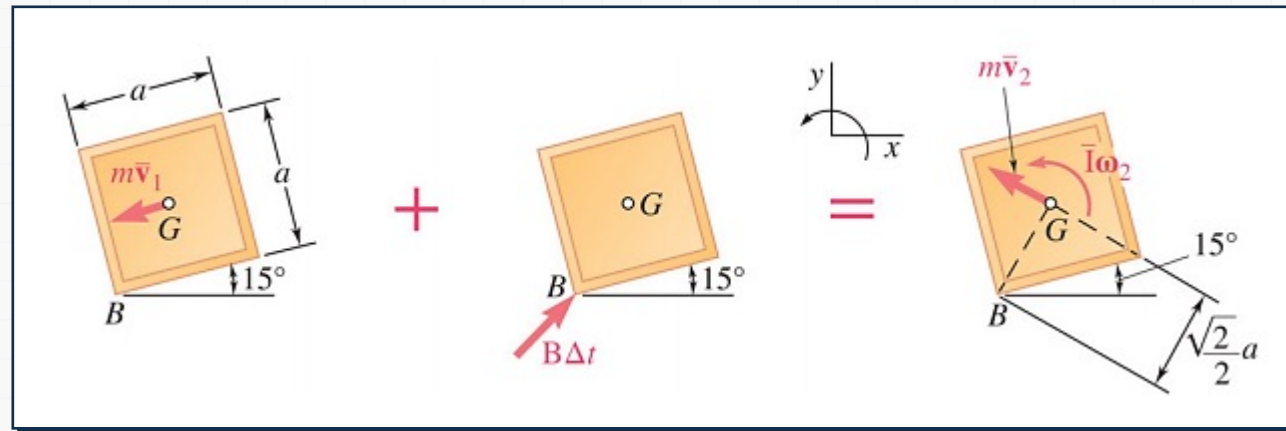
⚙ Modeling and Analysis

- Apply the principle of impulse and momentum to relate the velocity of the package on conveyor belt A before the impact at B to angular velocity about B after impact.

15 Impulse and momentum principle in planar motion

Sample Problem 6

⚙ Modeling and Analysis



+ ↺ moments about B

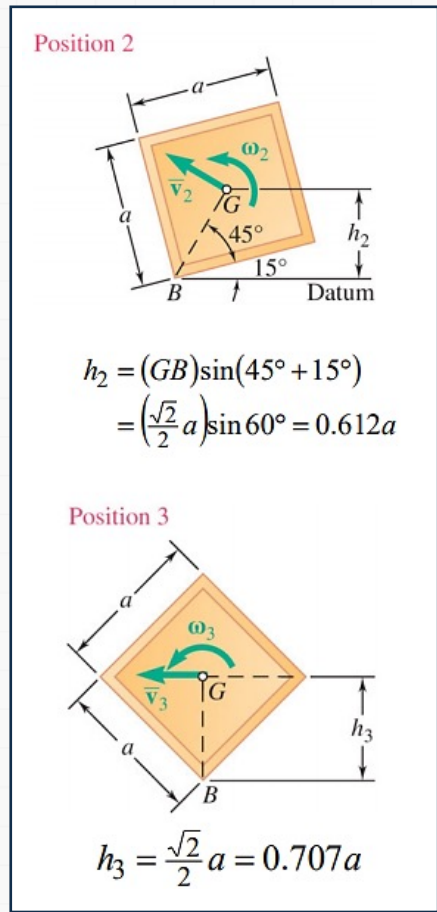
$$(m\bar{v}_1) \left(\frac{1}{2} a \right) + 0 = (m\bar{v}_2) \left(\frac{\sqrt{2}}{2} a \right) + \bar{I} \omega_2 \quad \bar{v}_2 = \left(\frac{\sqrt{2}}{2} a \right) \omega_2 \quad \bar{I} = \frac{1}{6} m a^2$$

$$(m\bar{v}_1) \left(\frac{1}{2} a \right) + 0 = m \left(\frac{\sqrt{2}}{2} a \omega_2 \right) \left(\frac{\sqrt{2}}{2} a \right) + \left(\frac{1}{6} m a^2 \right) \omega_2 \quad \bar{v}_1 = \frac{4}{3} a \omega_2$$

15 Impulse and momentum principle in planar motion

Sample Problem 6

⚙ Modeling and Analysis



- Apply the principle of conservation of energy to determine the minimum initial angular velocity such that the mass center of the package will reach a position directly above B .

$$T_2 + V_2 = T_3 + V_3$$

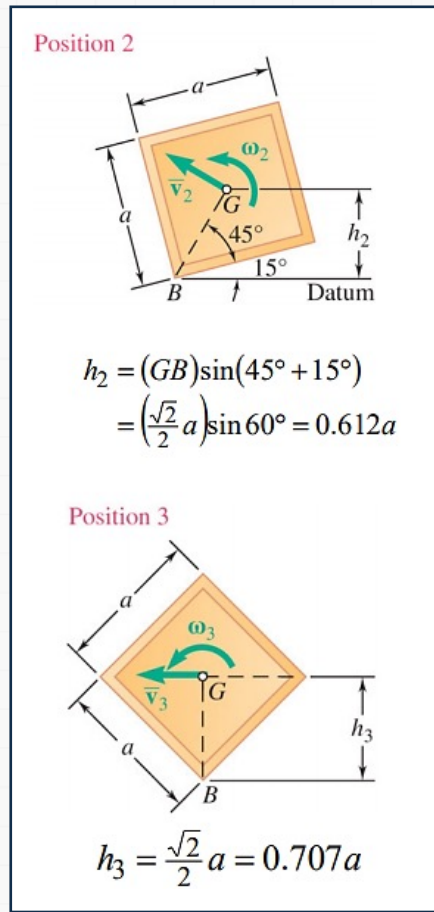
$$T_2 = \frac{1}{2}mv_2^2 + \frac{1}{2}\bar{I}\omega_2^2$$

$$= \frac{1}{2}m\left(\frac{\sqrt{2}}{2}a\omega_2\right)^2 + \frac{1}{2}\left(\frac{1}{6}ma^2\right)\omega_2^2 = \frac{1}{3}ma^2\omega_2^2$$

15 Impulse and momentum principle in planar motion

Sample Problem 6

Modeling and Analysis



$$V_2 = Wh_2$$

$$T_3 = 0 \quad (\text{Solving for the minimum } \omega_2)$$

$$V_3 = Wh_3$$

$$\frac{1}{3}ma^2\omega_2^2 + Wh_2 = 0 + Wh_3$$

$$\omega_2^2 = \frac{3W}{ma^2}(h_3 - h_2) = \frac{3g}{a^2}(0.707a - 0.612a) = \sqrt{0.285g/a}$$

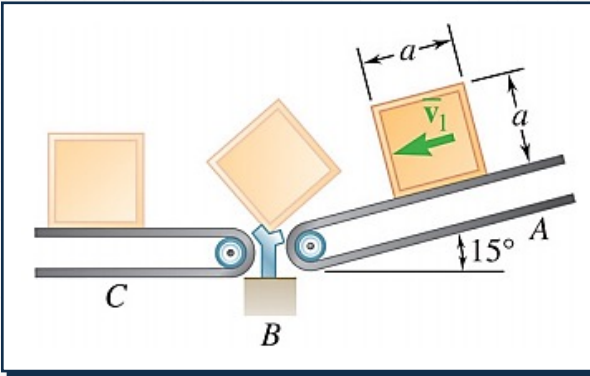
$$\bar{v}_1 = \frac{4}{3}a\omega_2 = \frac{4}{3}a\sqrt{0.285g/a}$$

$$\bar{v}_1 = 0.712\sqrt{ga}$$

15 Impulse and momentum principle in planar motion

Sample Problem 6

⚙ Modeling and Analysis

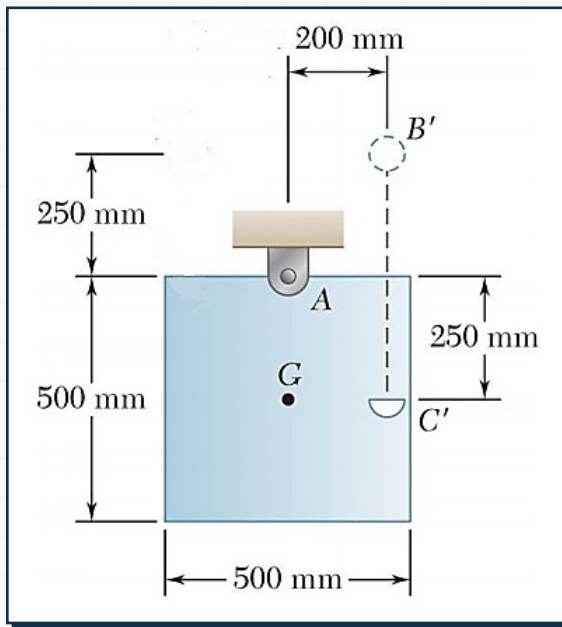


The combination of energy and momentum methods is typical of many design analyses. If you had been interested in determining the reaction at B immediately after the impact or at some other point in the motion, you would have needed to draw a **free-body diagram and kinetic diagram** and apply Newton's second law.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Determine the angular velocity



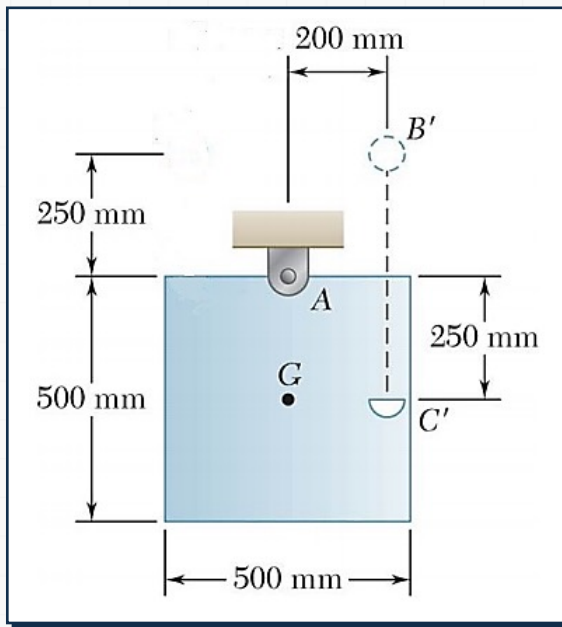
- An 8-kg wooden panel P is suspended from a pin support at A and is initially at rest. A 2-kg metal sphere S is released from rest at B' and falls into a hemispherical cup C' attached to the panel at the same level as the mass center G.

Assuming that the impact is perfectly plastic, determine **the angular velocity of the panel** immediately after the impact.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Determine the angular velocity



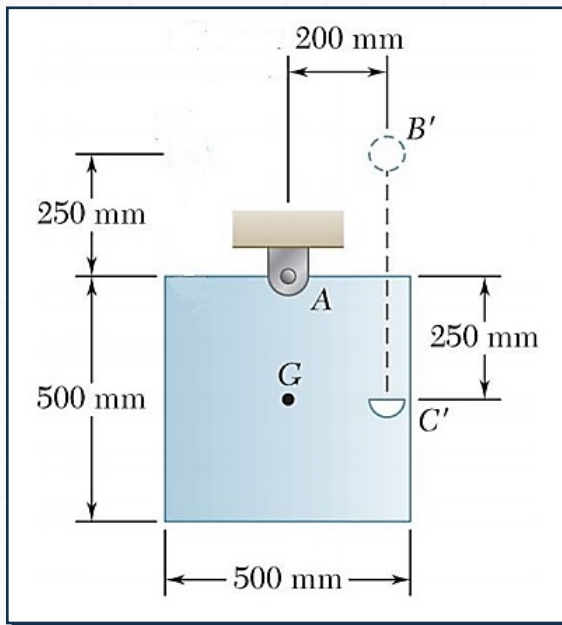
• Strategy

- Consider the sphere and panel as a single system. Apply the **principle of impulse and momentum**.
- The moments about A of the momenta and impulses provide a relation between the angular velocity of the panel and velocity of the sphere.
- Use the **principle of work-energy** to determine the angle through which the panel swings.

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



Given

$$\begin{aligned} m_s &= 2\text{ kg} & m_p &= 8\text{ kg} \\ h_s &= 0.250\text{ m} & e &= 0 \end{aligned}$$

Find

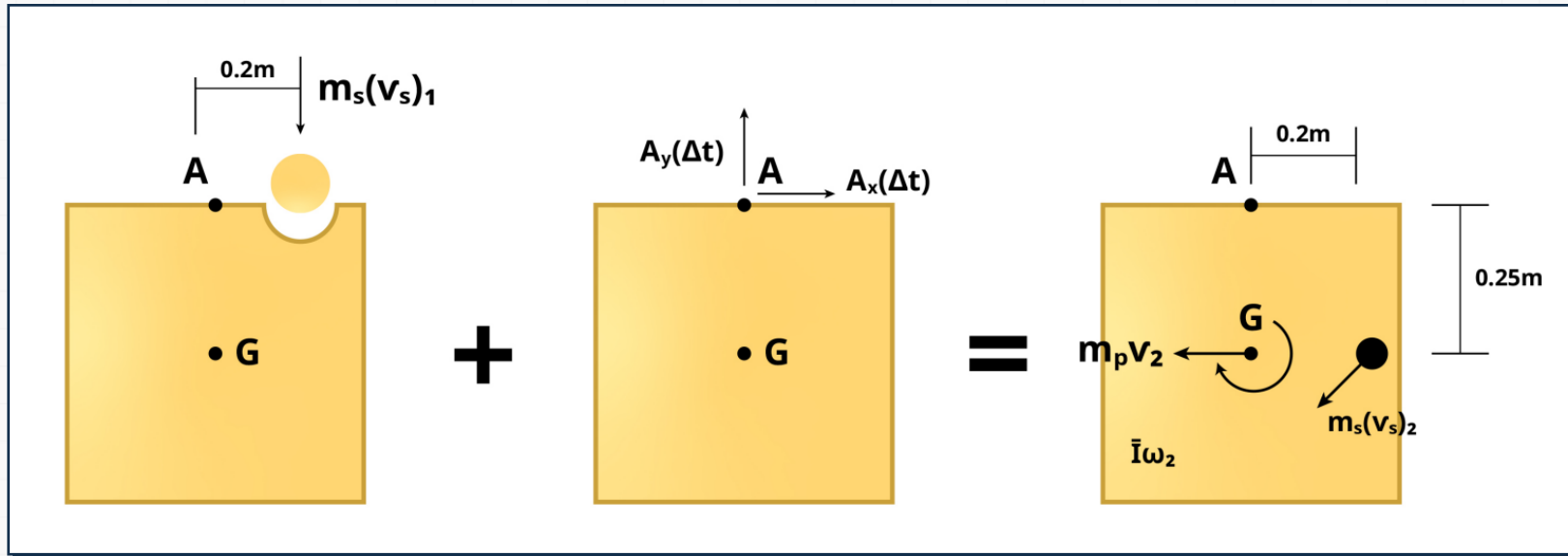
Angle θ through which the panel and sphere swing after the impact

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis

- Draw the impulse momentum diagram

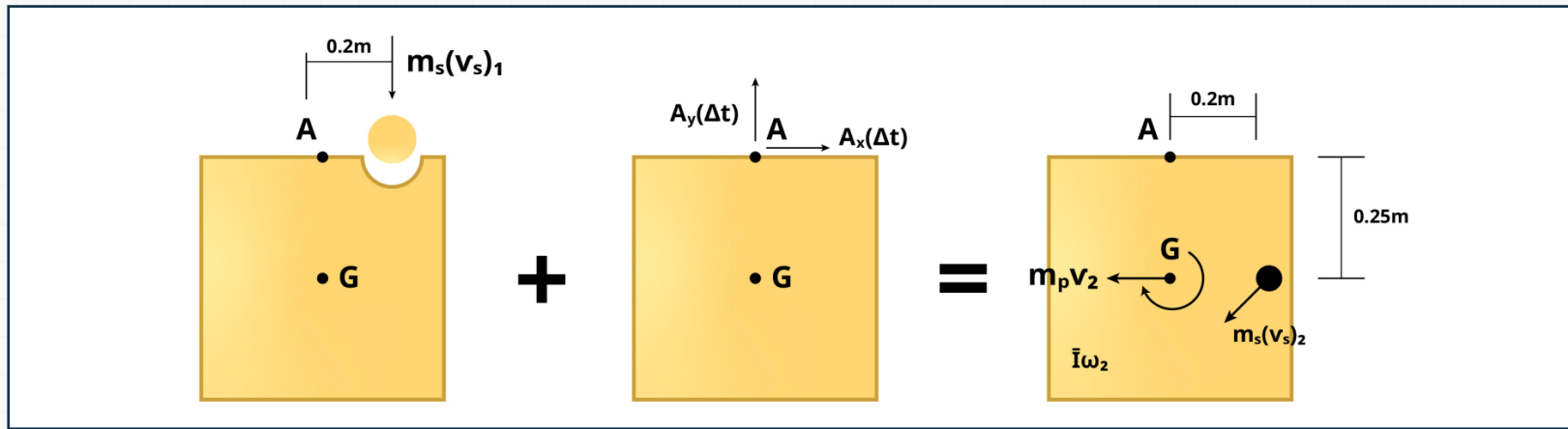


15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis

- Apply the angular impulse momentum equation about point A



$$\underline{m_s(v_{C'})_1(0.2)} + 0 = \underline{m_s(v_{C'})_2(AC')} + \underline{\bar{I}\omega_2 + m_p\bar{v}_2(0.25)}$$

H_A of sphere
before impact

H_A of sphere
after impact

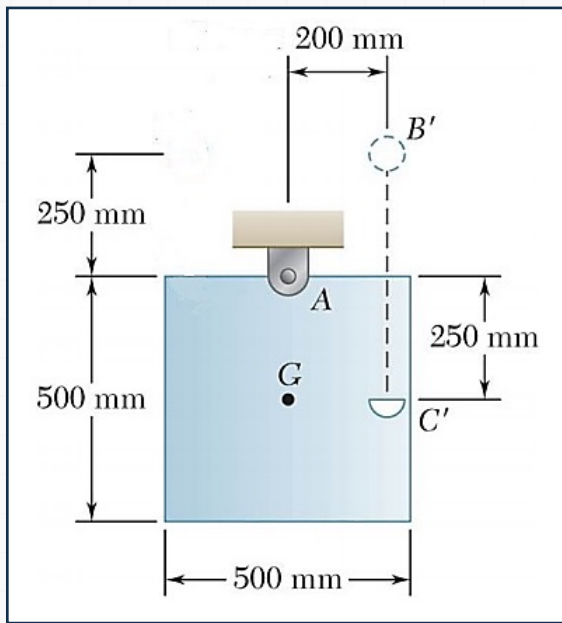
H_A of panel
before impact

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis

$$m_S(v_{C'})_1(0.2\text{m}) + 0 = m_S(v_{C'})_2(AC') + \bar{I}\omega_2 + m_P\bar{v}_2(0.25\text{m})$$



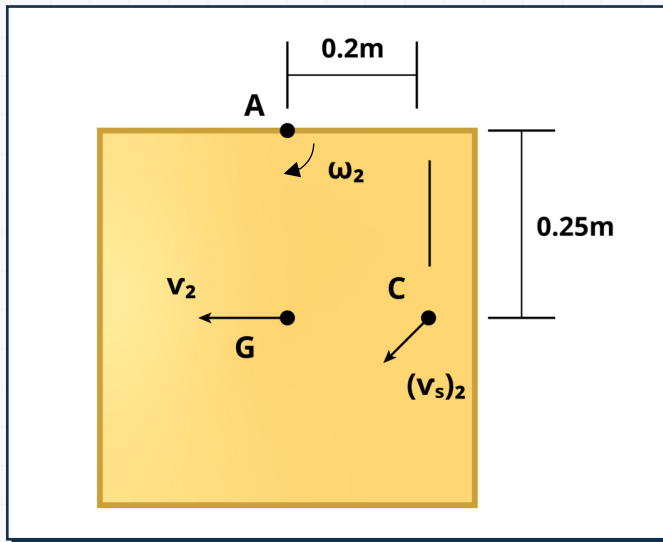
- Determine velocity of sphere at impact $(v_s)_1$
You can apply work-energy or kinematics.

$$\begin{aligned}(v_s)_1 &= \sqrt{2gy} \\ &= \sqrt{2(9.81)(0.5)} \\ &= 3.1321\text{m/s}\end{aligned}$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙️ Modeling and Analysis



- Determine velocity of sphere after impact in terms of ω_2

$$(\mathbf{v}_S)_2 = AC' \omega_2$$

$$AC' = \sqrt{(0.2)^2 + (0.25)^2} = 0.32016\text{m}$$

$$(\mathbf{v}_S)_2 = 0.32016\omega_2$$

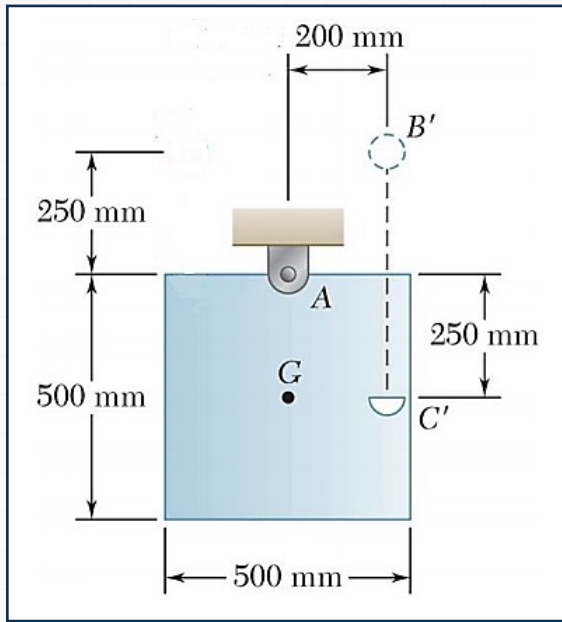
(perpendicular to AC)

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Group Problem Solving

⚙ Modeling and Analysis

$$m_S(v_{C'})_1(0.2) + 0 = m_S(v_{C'})_2(AC') + \bar{I}\omega_2 + m_P\bar{v}_2(0.25)$$



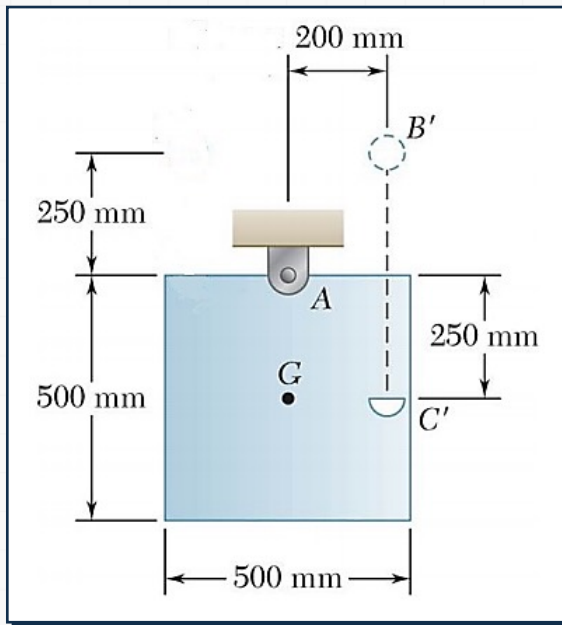
- Determine mass moment of inertia for panel

$$\bar{I} = \frac{1}{6}m_P(0.5)^2 = \frac{1}{6}(8)(0.5)^2 = 0.3333\text{kg} \cdot \text{m}^2$$

15 Impulse and momentum principle in planar motion

Group Problem Solving

⚙ Modeling and Analysis



- Substitute into H equation and solve for ω_2

$$m_S(v_{C'})_1(0.2) + 0 = m_S(v_{C'})_2(AC') + \bar{I}\omega_2 + m_P\bar{v}_2(0.25)$$

$$(2)(3.1321)(0.2) = (2)(0.32016\omega_2)(0.32016) + 0.3333\omega_2 + (8)(0.25)^2\omega_2$$

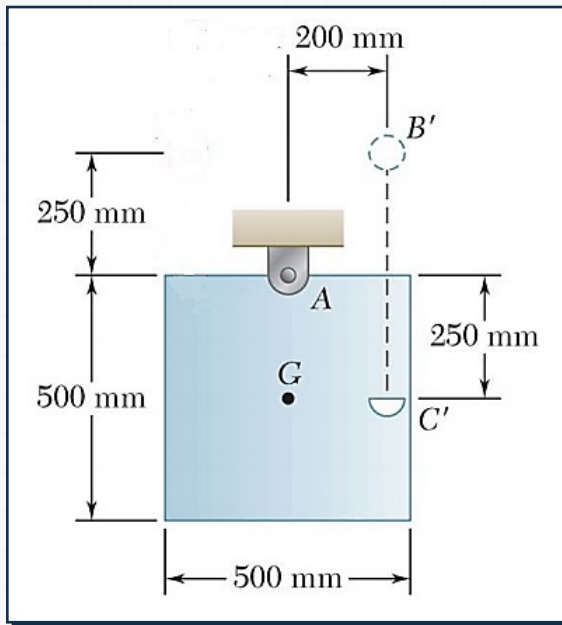
$$1.25284 = (0.2050 + 0.3333 + 0.500)\omega_2$$

$$\omega_2 = 1.207 \text{ rad/s}$$

15 Impulse and momentum principle in planar motion

Concept Question

⚙️ Determine how high up the panel swung after the impact



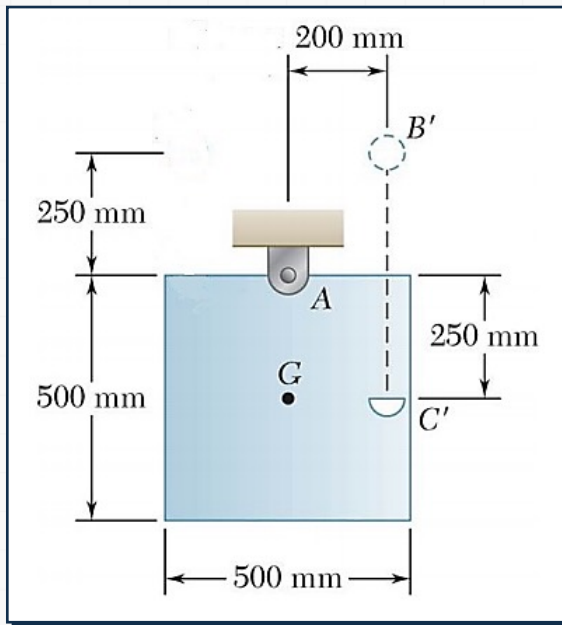
• For the previous problem, what would you do if you wanted to determine how high up the panel swung after the impact?

- a) Apply linear-momentum to solve for mv_G .
- b) Use work-energy and set T_{final} equal to zero.
- c) Use sum of forces and sum of moments equations.

15 Impulse and momentum principle in planar motion

Concept Question

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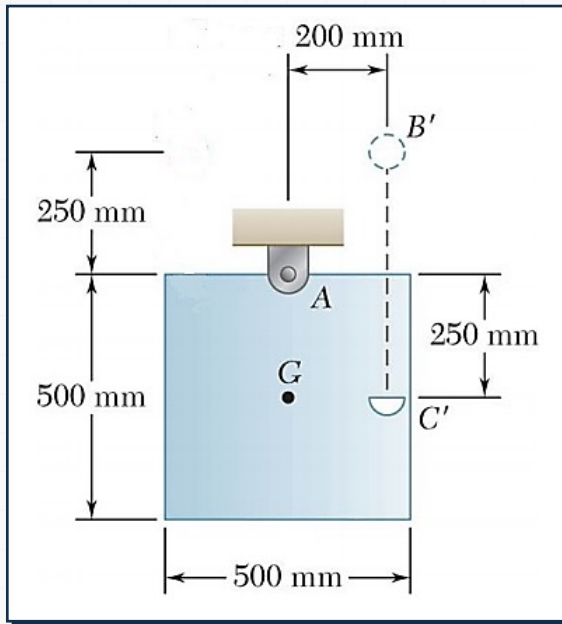


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15 Impulse and momentum principle in planar motion

Concept Question

⚙️ What if the ball was dropped closer to point A?



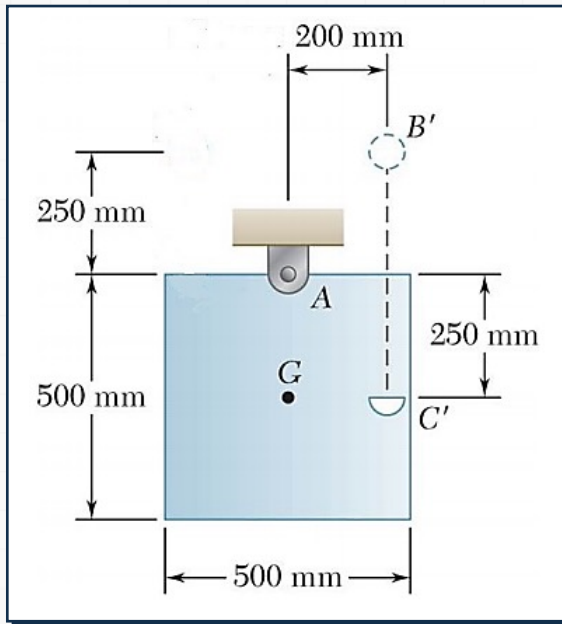
• For the previous problem, what if the ball was dropped closer to point A (example: at $x = 100\text{mm}$ instead of 200mm)?

- a) The angular velocity after impact would be bigger.
- b) The angular velocity after impact would be smaller.
- c) The angular velocity after impact would be the same.
- d) Not enough information to tell.

15 Impulse and momentum principle in planar motion

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15 Impulse and momentum principle in planar motion

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 17